

Democracy and Financial Intermediation: The Political Economy of Cryptocurrencies*

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Abstract

Cryptocurrency allows citizens to exit the formal financial system without leaving the state or the market. The exit choice runs along two dimensions: which currency instrument to hold (stablecoins or non-stablecoins) and which channel to use for exchange (intermediated or disintermediated venues). We argue that democratic institutions shape both dimensions. Where electoral accountability is strong, citizens remain in regulated intermediated channels; where it is weak, they shift toward dollar-pegged stablecoins routed through opaque venues, thereby reducing surveillance and currency debasement simultaneously. We build a model of democratic accountability and the cost(s) of financial legibility and derive hypotheses linking state power to exchange in cryptocurrency markets. We leverage unique transaction-level data from 110 emerging and developing economies, along with panel data and event study models, to test these hypotheses. Our findings show that behavior in cryptocurrency markets is directly tied to expectations regarding state power.

Introduction

In November 2020, the Central Bank of Nigeria abruptly froze the bank accounts of political activists protesting police misconduct (Amnesty International 2021). Following a coup attempt in 2016, the Turkish presidency issued emergency decrees that froze the assets of tens of thousands of citizens. In the months leading up to India’s 2024 election, that country’s Income Tax Department froze the accounts of the country’s largest opposition party (Reuters 2024). A decade ago, citizens facing such state actions would have had few options other than dollarizing: convert their holdings of domestic currency into foreign currency and hold those dollars in cash outside the financial system. This politically motivated currency substitution imposed a stark trade-off. Keeping deposits in the formal banking system preserved financial stability, but left account holders vulnerable to a state that had demonstrated its willingness to interfere in private finance. Exiting the formal banking system provided greater privacy but at a steep functional cost: an inability to conduct arm’s length transactions, store balances securely, or settle large payments.

Cryptocurrency dissolves this trade-off. Disintermediated markets operating outside the regulated financial system provide citizens with both an alternative architecture for transferring value and an alternative set of currencies to hold, without the functional sacrifices imposed by informal dollarization. The observed pattern is clear: in countries from Nigeria to Russia to Hong Kong,

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citizens route transactions through decentralized exchanges (software protocols that match buyers and sellers without an intermediary holding the assets) and peer-to-peer platforms (venues that connect users directly for bilateral exchange), accepting greater costs, lower liquidity, and the absence of consumer protection, ultimately paying more, not less, for this option.¹

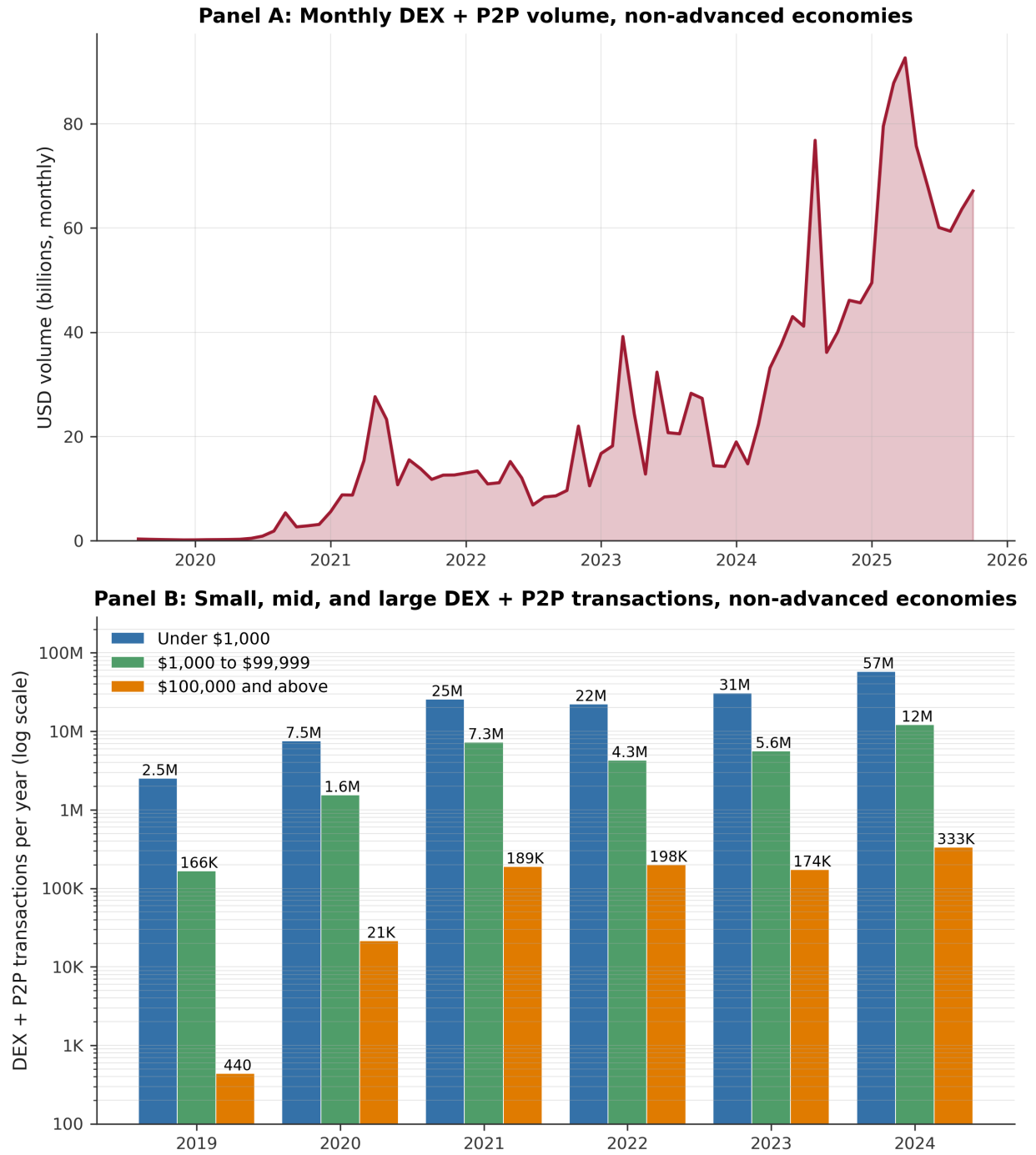
These are not isolated examples. Disintermediated cryptocurrency exchange markets, decentralized exchanges (DEX), and peer-to-peer (P2P) platforms have grown sharply over the past six years in both advanced and emerging economies. Annual retail DEX and P2P transaction counts in non-advanced economies rose from roughly 2.5 million in 2019 to 57 million in 2024, with mid-sized and large transactions growing on a comparable scale; advanced economies show the same trajectory (Figure 1 and Figure 2; Aramonte, Huang, and Schrimpf 2021; Chainalysis 2022).

Both the scale of these markets and their rapid growth are striking, signaling a shift in how citizens and states interact in monetary affairs. Traditionally, states make two distinct promises to their citizens about money: first, that currency issued today will retain its value; and second, that individuals can store wealth and transact without the financial system turning against them. Scholars have examined this first promise extensively, analyzing the institutional commitments— independent central banks, fixed exchange rates, currency boards, inflation targeting—that make it credible (Kyland and Prescott 1977; Barro and Gordon 1983; Broz 2002; Keefer and Stasavage 2003; Bodea and Hicks 2015). The second promise has received far less attention because, until recently, citizens had few options other than dollarization to bypass state-controlled financial systems. Disintermediated cryptocurrency markets remove that constraint: citizens can now use monetary instruments without routing them through intermediaries that the state can monitor, regulate, or compel.

We argue that democratic institutions limit the exploitation of the state’s financial observational capacity. When electoral accountability constrains the state’s ability to act on the information it collects, the political cost of being observed is low, and citizens are willing to use intermediated channels in exchange for the economic benefits those channels provide (Tsebelis 2002; North and Weingast 1989; Hollyer, Rosendorff, and Vreeland 2011). In the absence of political constraints, however, the state’s observational capacity becomes an instrument of surveillance, asset seizure, and political repression, prompting citizens to incur economic costs to escape it (Guriev and Treisman 2019; Escriba-Folch 2013; Cop 2020). The choice between intermediated and disintermediated exchange, therefore, is fundamentally a choice about how much citizens can afford to trust the institutions that govern financial exchange.

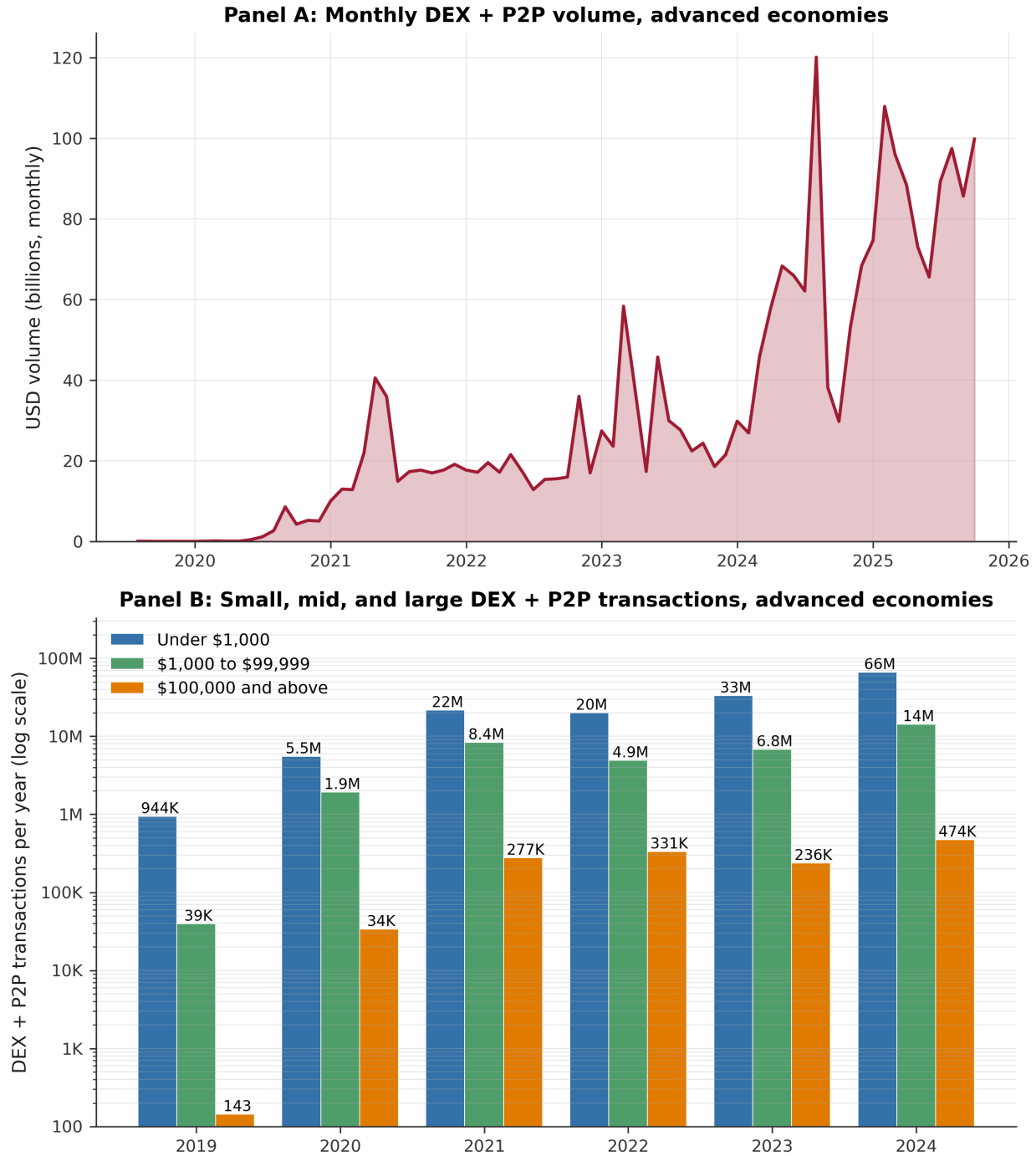
Our argument concerns two dimensions of cryptocurrency choice. On the instrument dimension, citizens under monetary stress demand dollar-denominated assets as a hedge against inflation and depreciation. On the channel dimension, citizens facing political oppression demand opaque exchange venues as a hedge against surveillance and asset seizure. Democratic accountability shapes both demands: where political institutions are strong, citizens trust both the currency and formal institutions; where both are weak, they turn to a parallel monetary system, holding dollar-denominated stablecoins and routing them through opaque channels. These patterns are not just theoretical; they are directly observable in the data. Figure 3 shows the composition of small-transaction USD

¹We measure the cost premium of disintermediated channels as the average across users who have functional access to the formal financial system. This premium is not universal. In countries where central bank exchange-rate management generates large spreads between official and parallel rates, peer-to-peer platforms can clear transactions at rates closer to the parallel market and may, ultimately, be cheaper for citizens transacting in the unofficial currency. Our argument requires that the political cost of legibility, rather than the gross transaction cost, drives the choice of financial architecture.



Source: Chainalysis. Top panel is USD volume. Bottom panel is annual transaction counts on a log scale.

Figure 1: Disintermediated cryptocurrency exchange in non-advanced economies, 2019–2024. Panel A: monthly USD volume routed through decentralized exchanges and peer-to-peer platforms. Panel B: annual transaction counts on the same venues, broken into three transaction-size bands (small under \$1,000; mid \$1,000 to \$99,999; large \$100,000 and above). Both panels are restricted to the 110-country non-advanced sample used in the empirical analysis.



Source: Chainalysis. Top panel is USD volume. Bottom panel is annual transaction counts on a log scale.

Figure 2: Disintermediated cryptocurrency exchange in advanced economies, 2019–2024. Panel structure and definitions identical to Figure 1. Advanced-economy retail and institutional activity track the non-advanced trajectory: small-transaction counts rise from roughly 944 thousand in 2019 to 66 million in 2024, and large-transaction counts from a few hundred to nearly half a million.

volume across the four cells defined by the cross-classification of instrument (stable vs. non-stable) and channel (intermediated vs. disintermediated) over our 2019–2024 window, and Figure 4 shows the same decomposition for large-transaction volume.

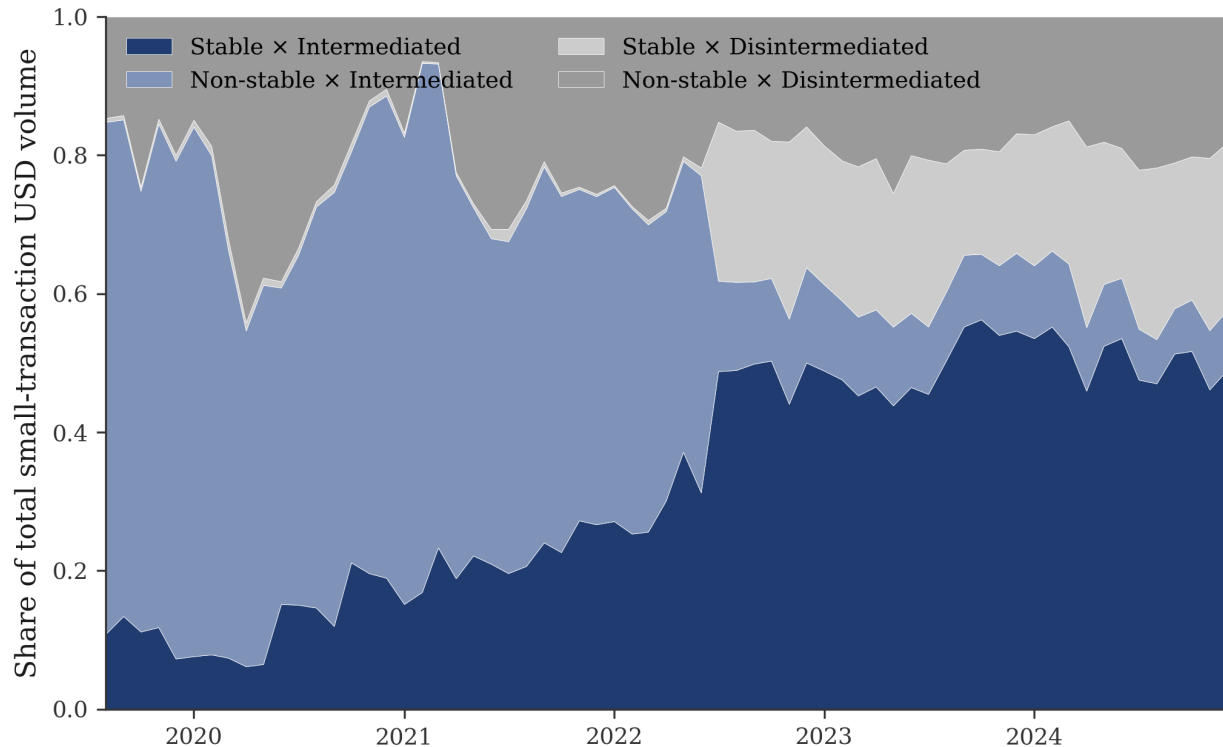


Figure 3: Four-cell composition of cryptocurrency activity, global sample (188 countries), small (retail) transactions, August 2019 through December 2024. Stacked shares of total small-transaction USD volume sum to one in each month. The four cells are defined by the crossing of stable versus non-stable instrument and intermediated versus disintermediated channel.

Fundamentally, we argue that democratic accountability constrains the executive’s use of state power against citizens. This claim has deep theoretical and empirical foundations (North and Weingast 1989; Scott 1998; Acemoglu and Robinson 2006; Hollyer, Rosendorff, and Vreeland 2011). From those foundations, we derive two testable claims. First, democratic accountability reduces the share of cryptocurrency activity routed through disintermediated channels, both in equilibrium and in response to discrete domestic political shocks. Second, the same legibility infrastructure that states use to monitor and discipline their own citizens is the instrument by which external governments project power across borders: sanctions compel correspondent banks to exit target jurisdictions, shutting regulated cross-border payments (Farrell and Newman 2019; Morse 2022). Sanctions add one further element: as they tighten, the availability of disintermediated channels degrades, producing non-linearity.

We test these hypotheses using data on cryptocurrency transactions from 110 emerging and developing economies between January 2019 and December 2024. For each country-month, we distinguish the share of crypto activity routed through intermediated channels, centralized exchanges that custody assets, verify identities, and report to regulators, from the share routed through disintermediated channels, which collect no identity data and transmit no records to state authorities. We first test the hypotheses using this cross-country panel and then examine market responses to

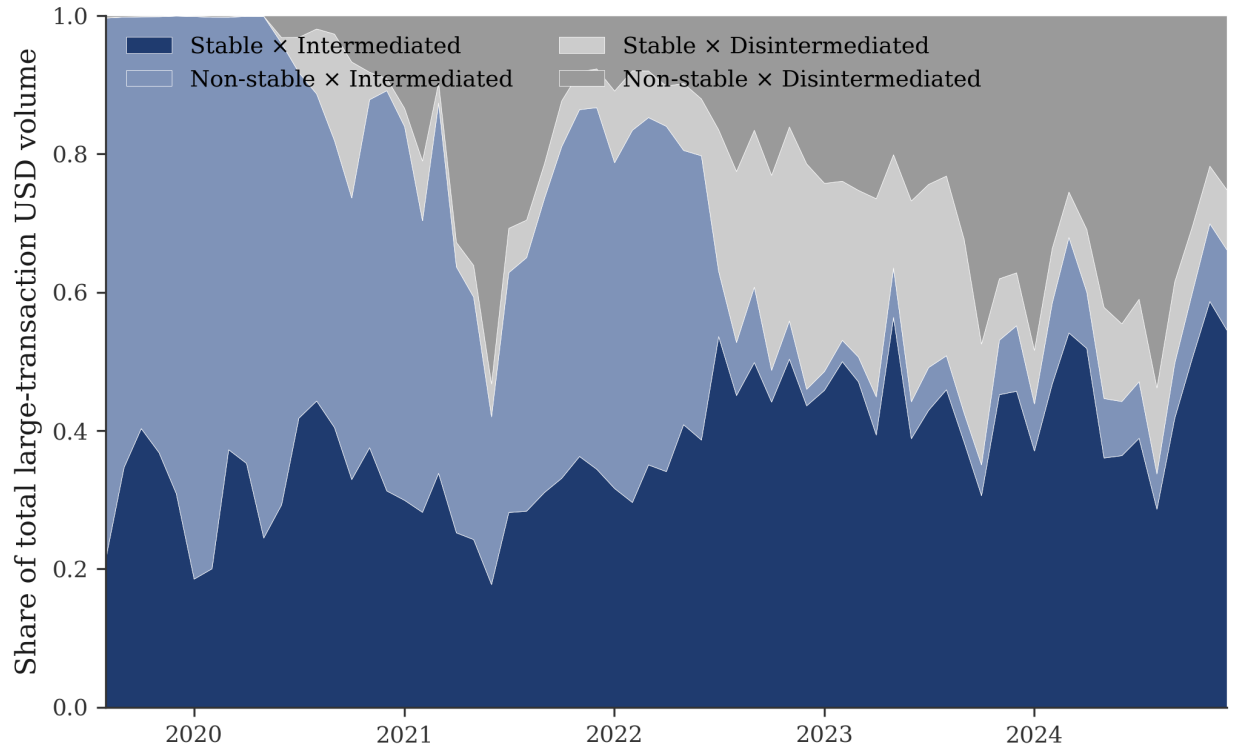


Figure 4: Four-cell composition of cryptocurrency activity, global sample (188 countries), large (institutional) transactions, August 2019 through December 2024. Stacked shares of total large-transaction USD volume sum to one in each month. Cell definitions are identical to Figure 3.

specific political and regulatory events that call into question the state’s commitment to neutrality in the financial system. We then analyze the effect of external pressure: international sanctions imposed on a target country’s banking system. In both cases, we find that internal and external restrictions increase the movement from intermediated to disintermediated markets for currency exchange.

Our arguments and evidence unfold as follows. Section I documents the historical consolidation of financial intermediation as state infrastructure and the resulting byproduct of legibility. Section II develops the political economy of financial legibility, the double-spending problem that blockchain-based cryptocurrencies are designed to solve, and the link to the choice between intermediated and disintermediated cryptocurrency channels. Section III builds a model in which democratic accountability operates on both the baseline political cost of legibility and on citizens’ adaptive capacity to substitute in response to external shocks. Section IV derives the hypotheses. Section V describes the data and empirical strategy. Section VI presents the empirical results: democratic accountability reduces disintermediation across countries and within countries after acute political shocks; international sanctions trace an inverted-U whose ascending portion democracy amplifies; and both findings hold in a combined model and under placebo tests. Section VII offers case evidence on the regulatory-crackdown mechanism through which states extend legibility infrastructure, and Section VIII concludes.

I. Money, Trust, and the Architecture of Visibility

Monetary exchange requires trust in the medium of payment, the counterparty, and the transaction record. Commodity money limits the need for trust but is impractical for routine settlement. As economies grew more complex, citizens therefore shifted to paper claims that could be transferred without physically exchanging the metal that backed them. The first financial intermediaries, goldsmiths and bullion-vault custodians, emerged to issue such claims against metal they held on deposit (Eichengreen 2019). Intermediation reduced the cost of solving the trust problem at scale: verifying identities, validating media of exchange, and maintaining records are costly when each party must perform these tasks independently, whereas an intermediary concentrates this work within a single institution (Diamond 1984; Catalini and Gans 2019).

This same concentration, however, created a distinct fragility. London’s goldsmiths discovered that they could lend out a portion of their reserves because depositors rarely demanded redemption simultaneously. The fractional-reserve banking system that followed functions only as long as depositors trust the intermediary. When that trust falters, depositors rush to withdraw, and the bank fails before it can liquidate the assets backing their deposits. The institutional response to recurrent bank runs has, in every case, been more intermediation: clearinghouses to manage counterparty risk, lender-of-last-resort facilities to provide emergency liquidity (Bagehot 1873; Gorton 1985; Goodhart 1988), and central banks placed atop this structure to supervise the system and backstop systemic failures.

Each stage in the evolution of financial institutions produces the same byproduct: legible records. The goldsmiths’ ledgers, the clearinghouses’ books, and the central banks’ supervisory data all document who holds wealth, in what amounts, and where — information the state can use to assess taxes and monitor economic activity (Scott 1998; Carruthers 1996).² Financial intermediation thus

²In 1640, Charles I seized £200,000 in bullion deposited at the Royal Mint, demonstrating that institutions

generates *legibility*: the state’s capacity to observe, identify, and link financial activity to specific legal persons. Legibility is thus an intrinsic feature of how the financial system operates rather than a deliberate design choice. The architecture of monetary trust and the architecture of state monitoring are, in this sense, two sides of the same coin.

Most accounts of state power over the economy emphasize the state’s interest in extracting seigniorage (Fischer 1982), securing macroeconomic autonomy through control of the money supply (Andrews 1994; Goodman and Pauly 1993), and using the payments system to monitor economic behavior (Cohen 1998; Helleiner 2003; Strange 1996). Each of these functions depends on financial legibility: states cannot tax, enforce regulatory compliance, or sanction political opponents without the capacity to observe how citizens use their money. The assembly of this legibility architecture was not accidental. War-making and fiscal capacity required that states obtain the ability to observe and tax economic activity at scale (Tilly 1990), and because direct monitoring of millions of citizens would be prohibitively costly, states learned over centuries to deputize a small number of intermediaries to monitor on their behalf (Levi 1988). The resulting infrastructure is expensive to construct but highly productive once established, and rulers with long time horizons have strong incentives to extend it (Olson 2000).

States operationalize this incentive through delegation. Rather than monitoring transactions directly, they license, regulate, and instruct private intermediaries to monitor on their behalf. Ayres and Braithwaite’s (1992) “enforced self-regulation” and the regulatory architecture that crystallized over the twentieth century are the institutional expressions of that delegation (Helleiner 1994). Domestically, the Bank Secrecy Act, Know-Your-Customer (KYC) mandates, and the Foreign Account Tax Compliance Act conscript banks as compliance agents reporting to the state.³ Internationally, SWIFT (1973) standardized state-legible interbank connections, and the Financial Action Task Force (1989) extended Anti-Money Laundering and Countering the Financing of Terrorism (AML/CFT) requirements globally through graylisting and blacklisting work similarly (Bütte and Mattli 2011; Sharman 2011; Simmons 2001; Morse 2022; McDowell 2016).⁴

By the early twenty-first century, virtually every non-trivial transaction in the formal financial system passed through a regulated intermediary that, by law, had to identify the parties, record the transaction, and make this record available to the state (Sharman 2011; Eichengreen 2019; Morse 2022). Money had become comprehensively legible, not because states observed every transaction directly, but because the intermediaries through which money moved were created, certified, and regulated as agents of state observation. This architecture enables taxation, financial repression, sanctions enforcement, and political surveillance in equal measure: the same intermediation that supports tax administration in Denmark facilitates financial repression in Egypt and the identification of dissidents in China.

consolidating private wealth were targets for sovereign extraction (Quinn 1997; Carruthers 1996). The goldsmiths’ private ledgers were only marginally safer than the Mint.

³The Foreign Account Tax Compliance Act (FATCA, 2010) extended American reporting requirements extraterritorially, requiring foreign financial institutions to report accounts held by U.S. citizens to the IRS (Grinberg 2012; Eccleston 2013). Each of these statutes is a state instruction to private intermediaries about what to collect, retain, and transmit, and each is unintelligible except as the formalization of the delegation Ayres and Braithwaite describe.

⁴FATF mutual evaluations are peer-review assessments of national AML/CFT regimes conducted on roughly seven-year cycles. Countries with significant deficiencies face graylisting or blacklisting, which has measurable consequences: correspondent banks reduce relationships with listed jurisdictions, and capital flows decline (Sharman 2011; Morse 2022).

II. The Political Economy of Financial Legibility

The political cost of financial transparency, the burden on citizens when their financial activities are visible to the state, depends on the strength of institutional limits on government power. In countries with strong democratic institutions, government actions concerning private wealth usually need judicial review and/or legislative oversight.⁵ Conversely, in places with weak institutions, the government can act on financial information with few obstacles. Significantly, all governments already have broad access to financial data; the key issue is whether existing institutional limits affect the costs of using that access.

Episodes in which political constraints fail or are suspended are evident in the shift of financial activity from intermediated to disintermediated markets. After Nigerian #EndSARS organizers had their bank accounts frozen in 2020, activity moved to peer-to-peer Bitcoin platforms (Amnesty International 2021). Following the 2020 National Security Law’s expansion of state powers to seize the assets of pro-democracy activists, Hong Kong residents rapidly adopted non-custodial wallets and decentralized exchanges. Russian dissidents and, after February 2022, ordinary Russians facing both Western sanctions and domestic asset freezes, shifted from regulated exchanges to peer-to-peer markets and decentralized protocols (Cop 2020). This pattern is not unique to autocracies: Canada froze the trucker convoy’s accounts in February 2022 but reversed course within weeks amid judicial review and parliamentary scrutiny.

Across regimes, what varies is the institutional cost of deploying financial coercion and, with it, the risk borne by a country’s citizens. A state that can monitor financial activity but is not credibly constrained from exploiting it generates sovereign risk: citizens discount the security of intermediated finance by the probability that the state will eventually turn the ledger against them.⁶ The extent of democratic accountability determines the size of that discount and thus whether financial transparency functions as a public good or as a surveillance instrument deployed against political opponents.⁷

This logic applies Hirschman’s (1970) exit–voice framework to monetary politics. When voice is available, citizens have little incentive to bear the costs of exit, and intermediated exchange prevails. When voice is unavailable, citizens who would otherwise prefer intermediated exchange accept the costs of disintermediation because the political cost of visibility exceeds the economic cost of opacity.⁸ The exit that cryptocurrency enables differs from earlier forms of dollarization. It

⁵Tsebelis (1995, 2002) formalizes this logic: policy change requires agreement among multiple actors with veto power, and the number and configuration of those actors determine the scope for executive discretion.

⁶The canonical formulation is North and Weingast’s (1989) account of the Glorious Revolution: the English Crown’s inability to credibly commit not to expropriate wealth was the fundamental obstacle to capital market development, and the institutional innovations of 1688 (parliamentary sovereignty, independent courts, secure property rights) solved the commitment problem by constraining the sovereign’s future action. Acemoglu and Robinson (2006) generalize this insight: a state that cannot bind itself to restraint must cede structural power if its commitments are to be credible.

⁷Two strands of literature speak directly to this point. Hollyer, Rosendorff, and Vreeland (2011) show that democratic transparency is two-sided: citizens monitor what the state does with the information it collects, and that monitoring constrains predatory use. Guriev and Treisman (2019, 2022) document the converse: financial surveillance, observable to the state but not to outsiders, is a useful instrument for identifying dissenters and punishing opponents without the visible coercion that traditional repression requires, thereby making it especially attractive to “informational autocrats.”

⁸This assumption does not require universal sophistication. Citizens facing acute political risk operate in dense networks within which channel-choice information propagates rapidly: after the 2020 #EndSARS account freezes, Nigerian organizers publicly disseminated instructions for receiving Bitcoin via peer-to-peer platforms; after February 2022, Russian opposition figures and independent media outlets did the same with non-custodial wallets. The channel-

is selective: citizens withdraw from one dimension of state authority, financial visibility, without exiting the currency or the political jurisdiction. It is graduated: transactions can be routed through state-visible channels for some purposes and state-invisible channels for others. And it is reversible: improvements in democratic accountability can draw citizens back into intermediated channels.

The risks of monetary debasement and financial surveillance tend to concentrate in the same countries because both originate from institutional weaknesses. Democratic institutions both enforce the commitment devices that make monetary stability credible and impose the constraints that make restraint in surveillance credible.⁹ A citizen who fears inflation can hedge through intermediated channels, for example, by buying dollars on a regulated exchange. A citizen who fears surveillance cannot do the same, because the exchange that provides access to dollars creates a record of the transaction. Cryptocurrency-based disintermediation addresses both risks at once.

Before 2008, intermediation was a technological necessity rather than a choice: every solution to digital double-spending required a trusted central authority to maintain a definitive ledger.¹⁰ That ledger-keeper was the chokepoint, in other words, the target for any state seeking to shut the system down or compel disclosure of its records. Nakamoto’s (2008) contribution solved the double-spending problem without a ledger-keeper by enabling a distributed network to reach consensus on transaction ordering through a cryptographic puzzle (“proof of work”) controlled by no single entity.¹¹

The digital currency ecosystem that has emerged, however, is not the one envisioned in Nakamoto’s paper. Centralized exchanges such as Mt. Gox, Coinbase, and Binance, along with a broader cohort of regulated platforms, rapidly reintroduced identity-verifying intermediation atop Bitcoin and the public blockchains that followed. Within a decade, states extended the same chokepoint strategy to these venues through IRS reporting guidance, the EU’s Fifth Anti-Money Laundering Directive, and the FATF Travel Rule. The resulting ecosystem is now bifurcated. On one side are regulated, KYC-compliant exchanges that are legible to the state. On the other side are decentralized exchanges, peer-to-peer platforms, and non-custodial wallets; a set of technologically diverse venues that do not transmit user identity to state authorities (Bailey, Rettler, and Warmke 2024). These disintermediated channels are built by citizens using parallel infrastructure because the formal infrastructure has been captured by states they cannot trust, rather than criminals seeking anonymity (Foley, Karlsen, and Putnins 2019).

The essential political feature of disintermediated markets is not anonymity but illegibility. At the level of the ledger, the Bitcoin blockchain is in fact the opposite of anonymous: every transaction is permanently recorded on a public ledger that anyone can inspect.¹² The ledger tracks wallet addresses rather than legal identities, and any link between an address and an individual must

choice signal also does not require formal regulatory understanding, only the recognition that some platforms ask for a passport photograph and others do not, an observation available to any user during account creation.

⁹The literature linking central bank independence and monetary credibility makes this institutional dependence explicit: Cukierman, Webb, and Neyapti (1992); Maxfield (1997); Broz (2002); Bodea and Hicks (2015).

¹⁰Double-spending is the problem that distinguishes digital from physical money. A physical banknote can be handed to one party at a time; a digital token, being a string of bits, can be copied and “spent” twice unless something prevents it. Prior digital-cash proposals (Chaum 1983; DigiCash; e-gold) solved this by having a central ledger-keeper record which tokens had already been spent and refuse to honor duplicates.

¹¹The technical details of the blockchain protocol (proof-of-work consensus, cryptographic hash functions, block structure) are described in Appendix A. The main text develops only the features necessary for the political theory.

¹²Forensic firms such as Chainalysis, Elliptic, and CipherTrace (contracted by the U.S. Treasury, DEA, FBI, and numerous foreign governments) have built substantial businesses on the underlying transparency of the blockchain. Their core technique is clustering: wallet addresses are grouped by transaction patterns, and each cluster is identified through external information.

be established outside the protocol. A citizen who acquires cryptocurrency through a centralized exchange has created that link in the exchange’s KYC system, and any authority able to compel the exchange can de-anonymize her entire transaction history. A citizen who acquires cryptocurrency through a peer-to-peer exchange with no KYC requirement has not.

Citizens in every country where cryptocurrencies are in active use, therefore, confront the same architectural choice: a legible channel that exposes them to whatever use the state makes of financial information, and an illegible one that does not. The cost they incur at this margin, the choice of intermediated versus disintermediated exchange, depends on their expectations that the state will refrain from predation. In the next section, we develop a model that yields two predictions: one concerning the equilibrium level of disintermediation and one concerning how citizens respond when external pressure raises the costs of exchange.

III. A model of Citizen Choice

In this section, we introduce a simple model formalizing the citizen’s choice between intermediated and disintermediated channels for currency transactions; that choice is governed by weighing the economic cost of using illegible cryptocurrency channels against the political cost of being observed in legible ones. We provide a simplified exposition here, leaving full derivations, sufficient conditions, and multi-tier and transaction-size extensions for Appendix B.

III.1 The Citizen’s Choice

A citizen decides how to route cryptocurrency transactions: through an intermediated channel that collects identifying information, or through a disintermediated channel that does not collect it. The political cost of legibility decomposes into two sources of risk: a domestic source, the probability that the citizen’s own state acts predatorily once it observes her, and an external source, the probability that an international sanctioning authority acts against her through the same legible channel. Both operate through the same mechanism: a regulated intermediary makes the citizen identifiable to whoever can compel its compliance, and the institutions that constrain the state moderate the risk.

Formally, the citizen chooses the disintermediated share of cryptocurrency holdings $(1 - \alpha) \in [0, 1]$ to maximize expected utility. The intermediated channel provides a non-price benefit $B > 0$ (consumer protection, deposit insurance, legal recourse). The disintermediated channel imposes a convex economic cost, $c_D^0/[2\psi(\delta)] \cdot (1 - \alpha)^2$, in which $c_D^0 > 0$ scales the level of the cost and $\psi(\delta) > 0$ captures the citizen’s institutional capacity to substitute, entering in the denominator so that greater institutional capacity lowers the effective cost of substitution. Higher democratic accountability raises this capacity, since $\psi'(\delta) > 0$. The political cost of legibility combines the domestic source $\pi(\delta) \in [0, 1]$, with $\pi'(\delta) < 0$, and an external source proportional to sanctions intensity S :

$$\Phi(\delta, r, S) = \pi(\delta) \cdot L(r) + \eta \cdot S,$$

where $L(r) > 0$ is the loss conditional on state predation. The citizen’s optimal mix is the point at which the political benefit of becoming less observable offsets the economic cost of routing around the intermediary and is represented by the following:

$$1 - \alpha^*(\delta, r, S) = \frac{\psi(\delta) \cdot [\Phi(\delta, r, S) - B]}{c_D^0}.$$

Disintermediation rises with the political cost of legibility Φ and with substitution capacity ψ ; it falls with the convex cost c_D^0 and with the value of intermediation B .

Two parameters carry the democracy effect, acting through different channels. $\pi(\delta)$ is the baseline-cost channel: democratic accountability lowers the probability that the domestic state acts predatorily on its observations, raising the relative value of legible channels and lowering the citizen's equilibrium disintermediated share. $\psi(\delta)$ is the adaptive-capacity channel: democratic accountability supplies the infrastructure that lowers the cost of substituting towards these channels (i.e., a free press, civil-society networks, a less censored internet, and weaker state capacity to suppress disintermediated activity through prosecution, account freezes, or platform blocks). The first sets the equilibrium level of disintermediation, including within-country adjustments to acute domestic shocks; the second sets the extent to which citizens move when external pressure raises the cost of remaining legible. Both express the same constraint on executive action, but they pull in opposite directions on the disintermediated share — democracy lowers its baseline through $\pi(\delta)$ and sharpens its response to external shocks through $\psi(\delta)$ — the tension that the sanctions analysis exploits.

III.2 Implications

Under normal conditions, the baseline-cost channel dominates the equilibrium share of disintermediated activity. Setting $S = 0$ reduces the political cost to its domestic term $\pi(\delta)L(r)$; differentiating the interior optimum with respect to democracy gives

$$\left. \frac{\partial(1 - \alpha^*)}{\partial \delta} \right|_{S=0} = \frac{\psi'(\delta) [\pi(\delta)L(r) - B] + \psi(\delta) \pi'(\delta) L(r)}{c_D^0}.$$

The two terms pull in opposite directions — the capacity term ($\psi' > 0$) is positive, the baseline-cost term ($\pi' < 0$) is negative — with the baseline-cost term dominating in the empirically relevant region (Appendix B.4 gives the sufficient condition). The equilibrium disintermediation share is therefore lower in democracies: democratic citizens accept intermediated channels because the economic benefits outweigh the small political cost, whereas citizens in autocracies bear the costs of disintermediation because the political cost of legibility is large.

For external shocks, the framework predicts a more complex pattern. Sanctions¹³ raise the political cost of legibility through the same mechanism as domestic predation, but external sanctions simultaneously degrade the disintermediated infrastructure itself:¹⁴ correspondent banking withdrawal,

¹³International financial sanctions are restrictions one state (or coalition) imposes on another's access to the global financial system — typically asset freezes on government and individual holdings, correspondent banking restrictions that force third-country banks to refuse transactions with the target, removal from SWIFT and other payment-messaging infrastructure, and secondary sanctions that penalize banks in non-sanctioning jurisdictions for dealing with sanctioned entities. Coalitions impose them in response to military aggression (Russia after February 2022), nuclear proliferation (Iran, North Korea), human rights violations (Myanmar, Belarus), and a range of other foreign-policy disputes.

¹⁴This supply-degradation channel is well documented. Faced with sanctions exposure, banks withdraw correspondent relationships from targeted jurisdictions to avoid compliance liability — the “de-risking” dynamic studied for sanctions and for FATF listing alike (Farrell and Newman 2019; Morse 2022; Sharman 2011). The same logic extends

exchange exit, and fiat on-ramp closure reduce the supply of channels through which citizens could exit. We capture this supply effect through a quadratic cost-shifter, $c_D^0(S) = c_D^{0,0} + \kappa S^2$, in which $\kappa > 0$ governs the convexity of infrastructure damage at high intensity. The disintermediated share at sanctions intensity S is

$$1 - \alpha^*(S) = \frac{\psi(\delta) \cdot [\pi(\delta)L(r) + \eta S - B]}{c_D^{0,0} + \kappa S^2}.$$

The denominator grows faster than the numerator at high S , generating an inverted-U whose peak satisfies

$$S^* = \frac{\eta \cdot c_D^0(S^*)}{2\kappa \cdot [\Phi - B]}.$$

An interior peak exists when κ is sufficiently large relative to η — equivalently, when supply-side correspondent-banking infrastructure damage rises faster than demand-side citizen substitution.

On the ascending portion of the inverted-U, where demand effects dominate and disintermediated infrastructure remains operational, the marginal effect of sanctions on the disintermediated share is approximately

$$\left. \frac{\partial(1 - \alpha^*)}{\partial S} \right|_{\text{demand region}} \approx \frac{\psi(\delta) \cdot \eta}{c_D^0}.$$

Differentiating again with respect to democracy yields:

$$\left. \frac{\partial^2(1 - \alpha^*)}{\partial S \partial \delta} \right|_{\text{demand region}} = \frac{\psi'(\delta) \cdot \eta}{c_D^0} > 0.$$

Democratic accountability *amplifies* the citizen’s response to a marginal sanctions increase, even as democratic accountability — through its effect on $\pi(\delta)$ in the baseline — *reduces* the equilibrium level of disintermediation. The amplification operates entirely through the adaptive-capacity term $\psi(\delta)$; on the descending portion of the inverted-U it vanishes, because correspondent banks do not condition their exit on the target country’s domestic political institutions. A framework operating on the political-cost level alone — lowering $\pi(\delta)$ but holding $\psi(\delta)$ fixed — would predict the opposite, that democracies respond *less* because they start from a lower baseline.

IV. Hypotheses

The model in Section III generates two primary hypotheses. First: democratic accountability reduces financial disintermediation, both in equilibrium and in response to acute domestic political

to the crypto exit channel: comprehensively sanctioned jurisdictions lose access to centralized exchanges and fiat on-ramps that convert local currency into cryptocurrency, as when major exchanges and card networks restricted Russian users after February 2022 and when access for Iranian users was entirely foreclosed. The quadratic form of the supply-cost shifter is an approximation to this process, as detailed in the Appendix.

shocks. The equilibrium claim here is that citizens in countries with higher democratic accountability conduct a larger share of cryptocurrency transactions through intermediated channels, all else equal. The shock-response claim states that when a country experiences a discrete political event that raises the threat of state predation — an emergency law, a court purge, or an invasion that triggers wartime governance — citizens shift toward less-observable routing channels in the months following the event. Both claims concern the share of transactions routed through state-legible infrastructure, not the level of cryptocurrency adoption.

From the discussion in Section II, we derive three implications of the first hypothesis. Implication 1a (legibility gradient). The democracy effect runs along the channels citizens choose for political reasons: positive at the identifying end and negative at the non-identifying end. Channels that identify the user to state authorities should show a positive coefficient, because citizens in democracies accept legibility and concentrate activity there; non-identity-transmitting channels should show the largest negative coefficient, because activity exiting state observation accumulates in them. The sign reversal across this identity-transmission threshold is a diagnostic feature — a sharp political threshold between identity-transmitting and non-identity-transmitting venues produces opposite-signed coefficients on adjacent points of the gradient. The model treats demand for the deepest-privacy and illicit-flagged venues as exogenous to political accountability (Appendix B.7); as such, it predicts no relationship with democracy in these venues. That null also serves as a falsification check: a democracy effect in venues used for non-political reasons would undercut the political-exit rationale we derive above.

Implication 1b (parallel prediction for stablecoins versus non-stablecoins). The same mechanism predicts an instrument-share pattern alongside the channel-share pattern: citizens facing weaker democratic accountability hold a larger share of their cryptocurrency in dollar-pegged stablecoins than in non-stablecoin assets. This implication is straightforward: democratic institutions embed commitment devices that enhance the credibility of monetary policy, just as they constrain politically motivated financial surveillance (Cukierman, Webb, and Neyapti 1992; Bodea and Hicks 2015).

Implication 1c (concentration in small retail transactions). The democracy effect is concentrated in the small-transaction subsample because individual citizens lack access to the institutional substitutes (e.g., offshore banking, OTC desks, corporate treasury management) that firms and multinationals use to hedge against domestic predation risk. For individuals, cryptocurrency-based disintermediation is the available substitute, so the political-cost mechanism is strongest where individuals transact.

Hypothesis 2: International financial sanctions have a non-linear effect on disintermediation, and democratic accountability amplifies citizens’ demand-side response in the sanctioned country. Sanctions enter through two channels that work in opposite directions. Moderate sanctions raise the political cost of legibility while disintermediated infrastructure remains operational, increasing the disintermediated share through citizen demand (Result 2 of Appendix B). Comprehensive sanctions degrade the infrastructure on which disintermediated channels themselves depend — correspondent banking, exchange access, fiat on-ramps — reducing disintermediation through supply. The net pattern is an inverted-U in sanctions intensity.

Implication 2a (steeper inverted-U for small transactions). Small transactions rely on local crypto infrastructure that comprehensive sanctions disproportionately degrade through correspondent banking withdrawal and exchange exits. Large transactions access a global infrastructure less sensitive to local disruption. The downturn at high sanctions intensity is therefore sharper for small trans-

actions.

Implication 2b (democracy amplifies the demand-side response). On the ascending portion of the inverted-U, democratic accountability raises citizens’ response to a marginal sanctions increase through the adaptive-capacity term $\psi(\delta)$ (Result 3 of Appendix B). This is the model’s signature claim. Democracy enters the disintermediation outcome through two parameters that pull in opposite directions: it lowers $\pi(\delta)$, the baseline political cost, reducing the equilibrium disintermediated share; it raises $\psi(\delta)$, the adaptive capacity, amplifying the marginal response to external shocks. On the descending portion, infrastructure destruction binds, and the amplification vanishes — correspondent banking withdrawal and protocol-level exit are independent of the target country’s domestic political institutions.

V. Data and Empirical Strategy

V.1 Data

We construct a country-month panel of 110 non-advanced economies from January 2019 through December 2024.¹⁵ Cryptocurrency outcomes come from Chainalysis transaction-level data, aggregated to monthly volumes by country, month, and legibility tier.¹⁶ We sort exchange channels into five tiers, ordered by the strength of the identity link they provide to state authorities. Tier 0 covers intermediated activity at licensed centralized exchanges, hosted wallets, and merchant services providers, which collect comprehensive KYC information at account opening and report transactions to the state. Tiers 1 and 2 cover venues with partial identity verification: lending protocols (Tier 1), smart-contract platforms, and no-KYC exchanges (Tier 2).¹⁷ Tier 3 covers decentralized exchanges, peer-to-peer marketplaces, and cryptocurrency ATMs, none of which transmit counterparty identity to the state. Tier 4 contains mixing services, protocol-level privacy tools, and residual activity from the darknet, sanctioned entities, and illicitly flagged activity.

Tier 3 and Tier 4 together sit on the unobservable side of legibility, where the state cannot, without external information, link a transaction to an identified citizen. They differ in why citizens transact there. Tier 3 activity is consistent with the political-cost mechanism: citizens choosing channels that do not identify them to their state. Tier 4 activity is dominated by non-political demand (e.g., gambling, criminal mixing, and sanctioned-entity flows) that do not, ex ante, respond systematically to domestic institutional conditions. Together, they form the deep-disintermediation region.

¹⁵We classify countries as non-advanced using the IMF *World Economic Outlook* Advanced Economy / Emerging Market and Developing Economy (AE/EMDE) typology in its 2024 vintage. The Chainalysis database contains 187 countries with attributable transaction volume in the 2019m8–2024m12 window, of which 146 are non-advanced under the IMF classification. We retain countries with at least 53 months (roughly 80 percent) of non-missing tier-share data over the 65-month window, which excludes 36 sparsely covered country panels, yielding a working sample of 110 countries and 6,911 country-months. Argentina and Türkiye both qualify as non-advanced and are retained.

¹⁶Chainalysis identifies wallets, clusters them into entities using transaction-pattern heuristics, and attributes activity to countries using IP information at fiat on-ramp interactions and KYC data shared by cooperating exchanges. The data, therefore, identify country-level activity with measurement error that increases for users routing transactions through privacy-preserving tools and through jurisdictions with weak KYC enforcement. The measurement error runs counter to our hypotheses: countries where citizens conceal their activity from intermediaries are countries where Chainalysis attribution is least reliable, biasing the disintermediated share downward in the countries where the political-cost mechanism predicts it should be highest. The bias works against finding the patterns we report.

¹⁷Although no-KYC exchanges do not require identity documents at signup, they remain centralized operators that accumulate identifying metadata (email addresses, IPs, device fingerprints, internal address-to-account links accessible to the state through legal or coercive means. This distinguishes them structurally from Tier 3 venues, where no custodial operator exists to compel the disclosure of information.

Institutional variables come from V-Dem v14, with V-Dem’s Clean Elections Index (`v2xel_frefair`) as our primary measure of democratic accountability. Macroeconomic controls — the real interest rate and exchange-rate depreciation — are drawn from the IMF’s International Financial Statistics. We include a measure of capital controls using the IMF’s Financial Account Restrictiveness Index (Fernández et al. 2016). International sanctions intensity is measured by the Global Sanctions Database (GSDB v4) as the log of 1 plus the number of countries imposing financial sanctions on the target. Regulatory variables — banking restrictions, payment bans, and FATF grey-listing — are coded from primary regulatory documents.

Every specification includes a block of crypto-market and global-financial controls that capture the common conditions driving activity across all countries: the price deviation of USDC (a US-dollar pegged stablecoin) from its dollar peg, the lagged Ethereum network gas price, the broad dollar index, and an indicator of centralized-exchange distress episodes; we include, separately, an indicator for China’s crypto ban capturing the People’s Bank of China’s blanket prohibition on cryptocurrency trading and mining from September 2021 through the end of the sample.

V.2 Dependent Variables

The theoretical model treats α as the country-aggregate share of cryptocurrency activity routed through intermediated channels; empirically, we work with the disintermediated share ($1 - \alpha$). Our primary outcome is the deep-disintermediation region defined above — the combined Tier 3 and Tier 4 share — which we construct in three variants: the small-transaction share, the all-transaction share, and the large-transaction share.

For the gradient analysis of Implication 1a, we construct the tier-share variables `tier0_share` through `tier4_share`, along with small-transaction variants. For the instrument channel, we use the share of USDC in aggregate holdings, computed as `USDC volume / (USDC volume + Bitcoin volume)`, capturing the dollar-substitution margin that operates alongside the channel dimension.

In the within-country event studies (Section VI.2), we use the volume-weighted channel illegibility index for small (under \$1,000) stablecoin transactions, denoted $\hat{L}^{\text{stable}}$. It is the volume-weighted average of the legibility weights τ_k across Chainalysis service categories, with weights ranging from 0 (centralized exchange) to 1 (darknet markets and sanctioned entities).¹⁸ We construct an analogous index over small non-stablecoin transactions, denoted $\hat{L}^{\text{nonstable}}$. The layered-evasion gap $\hat{L}^{\text{stable}} - \hat{L}^{\text{nonstable}}$ is the difference between the two indices: a positive value means citizens route their stablecoin volume through less-observable channels than their non-stablecoin volume — stacking two evasion strategies (a dollar-pegged instrument and an illegible routing channel) on the same volume rather than relying on either alone.

¹⁸The legibility weight τ_k is assigned to each Chainalysis service category k based on the strength of the identity link the venue transmits to the state authority. The mapping is as follows. Tier 0 ($\tau = 0.0$): centralized exchange, hosted wallet, merchant services. Tier 1 ($\tau = 0.2$): lending, lending contract. Tier 2 ($\tau = 0.4\text{--}0.5$): smart contract, token smart contract, ICO, NFT platform collection (each 0.4); no-KYC exchange (0.5). Tier 3 ($\tau = 0.6\text{--}0.7$): decentralized exchange, decentralized exchange contract, bridge (each 0.6); peer-to-peer exchange, ATM (each 0.7). Tier 4 ($\tau = 0.9\text{--}1.0$): mixing, protocol privacy (each 0.9); darknet market, sanctioned entity, sanctioned jurisdiction, sanctions, illicit actor/organization, fraud shop, scam, child abuse material (each 1.0). The mining pool, infrastructure-as-a-service, unclassifiable “other” volume, and the gambling service category are excluded from gradient construction. The gambling exclusion follows from per-category diagnostics reported in Section VI.1.

V.3 Estimation

The variation that identifies Hypothesis 1 is between countries, not within them. Democratic accountability is a slow-moving institutional feature that barely changes within a country over a six-year panel; the conventional approach, absorbing unobserved country-specific heterogeneity, country fixed effects, would discard almost all the variation that identifies the independent variable. We therefore estimate the panel data models using correlated random effects (CRE), implemented via the Mundlak transformation, in which the country-level mean of each time-varying regressor is included as an additional control (Mundlak 1978; Wooldridge 2010, ch. 10). CRE retains the between-country variation that carries the signal while purging the confounding that fixed effects are meant to remove.¹⁹

Year fixed effects capture global cryptocurrency-market shocks. We report standard errors clustered at the country level.

VI. Results

VI.1 Hypothesis 1: Democratic Accountability and Equilibrium Channel Choice

Table 1 tests Hypothesis 1 across three related outcomes: aggregate legibility, deep disintermediation, and deep disintermediation in small transactions (full specifications for all models are in the appendix). The Clean Elections Index coefficient is negative and statistically significant in each column. Moving from full autocracy to full democracy is associated with roughly a 10–percentage point reduction in the equilibrium disintermediated share in the small-transaction subsample, where we argue the political-cost mechanism is concentrated; the coefficient for all transactions is smaller in absolute magnitude but has the same sign.

Table 1. Democratic accountability is associated with lower equilibrium disintermediation.

Variable	Legibility Score	Deep Disint (All)	Deep Disint (Small)
Clean Elections Index	−0.0493* (0.0263)	−0.0630* (0.0366)	−0.1007** (0.0398)
N	4,242	4,242	4,242

*CRE estimator with year fixed effects; country-clustered standard errors in parentheses. Coefficients on the Clean Elections Index only; control coefficients are suppressed for readability and reported in full in Appendix E.1. Controls: lagged real interest rate, capital-account restrictiveness (FAR), exchange-rate depreciation, FATF grey-list, USDC price deviation, China crypto-ban indicator, lagged Ethereum gas price, centralized-exchange distress indicator, and the broad dollar index, plus Mundlak between-country means. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

Table 1 reports the average effect of democracy across all legibility tiers. Implication 1a makes a sharper prediction: when we disaggregate across tiers, the democracy coefficient should reverse

¹⁹Two properties make the estimator appropriate here. First, adding the country average of each time-varying regressor purges between-country confounding as unit fixed effects would, but without discarding the substantive between-country variation; under the Mundlak assumption, it is consistent and weakly more efficient than within fixed-effects estimation when between-unit variation carries the identifying signal. Second, it nests a direct test of the random-effects assumption: the joint significance of the Mundlak means.

sign as we move from fully legible to fully illegible venues. The pattern in Figure 5 matches the framework. Tier 0, exchange in fully intermediated markets, has a positive and significant coefficient: citizens in democracies accept the requirements of legibility and concentrate exchange in licensed venues. The opposite holds in Tier 3, where citizens in lower-accountability countries move out of identity-transmitting exchanges. The middle tiers, venues with only partial identity verification, sit tightly around zero, and that flat middle is the signature of a threshold response. If citizens were responding to legibility by degree, the coefficient would steadily become more negative tier by tier; instead, the effect remains near zero until identity transmission stops altogether, then turns sharply negative at Tier 3. Citizens are not fine-tuning to how observable a venue is — they exit once a venue stops reporting their identity to the state. Tier 4 is null, matching the model’s treatment of Tier 4 demand as exogenous to political-cost selection.²⁰

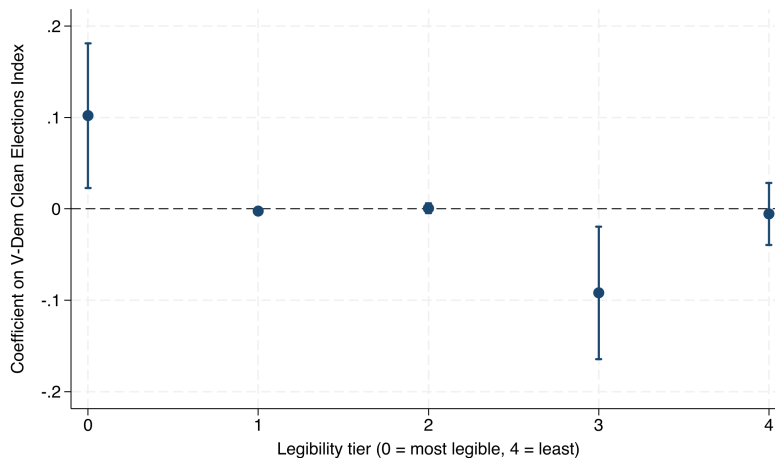


Figure 5: Democracy effect across legibility tiers, small-transaction outcomes. CRE estimator with year fixed effects. Coefficient on Clean Elections Index from separate regressions on each tier share `tier{0..4}_share_sm`. Vertical bars are 95% confidence intervals.

These results test the relationship between democracy and how citizens route transactions. But citizens must also decide what to hold. The relevant instrument is the stablecoin, a cryptocurrency token pegged to the US dollar that preserves value when the domestic currency is unstable and formal dollar accounts are hard to obtain. The second implication of H1 is that democracy operates

²⁰The Tier 4 category contains mixing, protocol-privacy, and illicit-flagged services (darknet market, sanctioned entity, sanctioned jurisdiction, illicit-actor-organization, fraud shop, scam, and child-abuse material). The pooling assumption is that demand for these services is driven primarily by non-political motives. Citizens with strong political-cost concerns might, in principle, use mixing services, thereby contaminating the null prediction at Tier 4. We disaggregate Tier 4 into two subcomponents from the raw Chainalysis service-category labels: a privacy-tool subcomponent (mixing plus protocol-privacy), which is the component a politically motivated user would plausibly select, and an illicit residual (darknet market, sanctioned entity, sanctioned jurisdiction, illicit-actor-organization, fraud shop, and scam). Both subcomponents produce null Clean Elections Index coefficients: the privacy subcomponent at $b = -0.006$ ($SE = 0.005$, $p = 0.18$) and the illicit residual at $b = +0.006$ ($SE = 0.021$, $p = 0.79$). The pooled Tier 4 null reported in Figure 5 thus reflects null effects for each subcomponent rather than a combination of oppositely signed components. The “gambling” service category was removed from the Tier 4 construction because per-category diagnostics on the non-AE sample show that gambling concentrates in offshore financial centers, FX-controlled jurisdictions, and weak-state autocracies rather than in regulated gambling hubs; co-moves negatively with centralized-exchange volume (Pearson -0.27) rather than with the privacy or illicit categories; and captures 41 percent of its USD volume in transactions above \$100,000, all of which are inconsistent with the retail political-cost mechanism Tier 4 is designed to identify. Gambling is excluded from the gradient by the same logic that excludes mining-pool, infrastructure-as-a-service, and “other” volume.

on this instrument margin, where accountability is weaker and monetary commitments less credible, and citizens move a larger share of their holdings into stablecoins. Table 2 bears this out. The Clean Elections Index is negative on both the USDC share (Column 1) and the deep-disintermediation share (Column 2), the channel effect being the larger of the two. Citizens in more democratic countries allocate a smaller share of their portfolios to dollar-pegged stablecoins, consistent with lower monetary-stress demand for dollar hedging, while the exchange-channel effect reflects the previously documented political-exit dimension. Democratic accountability thus constrains two forms of state extraction simultaneously: monetary debasement and political predation.

Table 2. Democracy operates on both instrument and channel margins.

Variable	USDC Share	Deep Disint (Small)	Stable Legibility	Nonstable Legibility
Clean Elections Index	−0.0528*	−0.1007**	−0.0217	−0.0542*
	(0.0317)	(0.0398)	(0.0167)	(0.0298)
N	4,242	4,242	4,221	4,242

*Same specification and controls as Table 1 (Clean Elections Index shown; full models in Appendix E.2). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

The third implication of H1 is that the democracy effect will be concentrated in small retail transactions, because individuals lack the institutional substitutes such as offshore banking, OTC desks, and corporate treasuries that firms and multinationals use to hedge against government predation. Table 3 confirms this hypothesis: the effect is largest in the small-transaction column and smaller in the all- and large-transaction columns.

Table 3. Democracy effect concentrates in small transactions.

Variable	Small	All	Large
Clean Elections Index	−0.1007**	−0.0630*	−0.0551
	(0.0398)	(0.0366)	(0.0421)
N	4,242	4,242	4,209

Same specifications and controls as Table 1 (full models in Appendix E.1).

It is important to note that the Clean Elections Index is a composite proxy for two theoretical mechanisms: it measures both electoral integrity — the degree to which incumbents face binding electoral accountability — which the framework links to both the baseline-cost channel $\pi(\delta)$ (accountable executives are less able to act on what they observe) and the adaptive-capacity channel $\psi(\delta)$ (electoral accountability travels with the free press, civil-society networks, and open internet that lower the cost of substitution). A single index cannot separate the two channels on its own, so we turn to auxiliary controls and report those results in Appendix D. Adding a battery of citizen-capacity and remittance-corridor controls — internet penetration, fixed broadband subscriptions, mobile cellular subscriptions, electricity access, log GDP per capita, and remittance intensity — absorbs the $\psi(\delta)$ infrastructure and leaves the democracy coefficient essentially unchanged across every specification (Appendix D). The surviving democracy effect is identified from the baseline-cost channel $\pi(\delta)$ rather than from raw substitution capacity. The small-transaction result is also

robust to measuring democracy using the V-Dem polyarchy index instead of the Clean Elections Index (Appendix D.6), the only alternative democracy index available in our panel.

VI.2 Hypothesis 1: Within-Country Responses to Acute Political Shocks

Beyond a cross-sectional response, Hypothesis 1 predicts within-country responses to discrete institutional shocks. We test this in two ways. First, we examine Hong Kong’s imposition of the National Security Law on June 30, 2020, Russia’s exclusion from the SWIFT payment-messaging system announced on February 26, 2022, and two regulatory changes in Nigeria. These cases represent sharply timed shocks that, under Hypothesis 1, should generate marked increases in $\hat{L}^{\text{stable}}$ within months. Second, we broaden beyond these named cases to the full set of political transitions cataloged in the V-Dem Episodes of Regime Transformation dataset (ERT). The ERT is a V-Dem companion that algorithmically identifies sustained movements in a country’s level of democracy, flagging each episode of autocratization, democratization, or regime breakdown, along with the year in which it begins. Because V-Dem codes episodes at an annual resolution while our panel runs monthly, we hand-code the treatment month of each onset from primary news and policy sources (Appendix C gives the dating procedure and cohort list). These episodes begin in different months across countries, so the treatment is staggered, a setting in which conventional two-way fixed-effects estimators are biased when treatment effects are heterogeneous. We therefore estimate the effect of these shocks on channel illegibility (transactions under \$1,000) with the Callaway–Sant’Anna estimator for staggered, heterogeneous-cohort difference-in-differences (Callaway and Sant’Anna 2021), implemented with a doubly robust, augmented inverse-probability weight.

Table 4 reports the post-period average treatment effects for the single-country cases (Hong Kong, Russia, and the two Nigerian events discussed below), and Figure 6 plots the dynamic event-study profiles for Hong Kong and Russia.

Table 4. Single-case event studies for flagship acute political shocks.

Case	Event	Date	Window	$\hat{L}^{\text{stable}}$	$\hat{L}^{\text{small_agg}}$
Hong Kong	National Security Law	2020-06	0–48	+0.099***	+0.130***
Russia	SWIFT exclusion	2022-02	0–34	+0.072***	+0.057***
Nigeria	#EndSARS account freezes	2020-11	0–24	not significant	+0.034**
Nigeria	#EndSARS, full window (with confounds)	2020-11	0–49	+0.024***	+0.043**
Nigeria	Central Bank crypto-banking ban	2021-02	0–47	not significant	not significant

*Single-treated-unit Callaway–Sant’Anna doubly-robust difference-in-differences with country-clustered standard errors and a never-treated donor pool (non-advanced economies with no ERT-coded political-shock onset in 2020–2024). The Nigeria-EndSARS specification is reported in two post-treatment windows, discussed below. $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.**

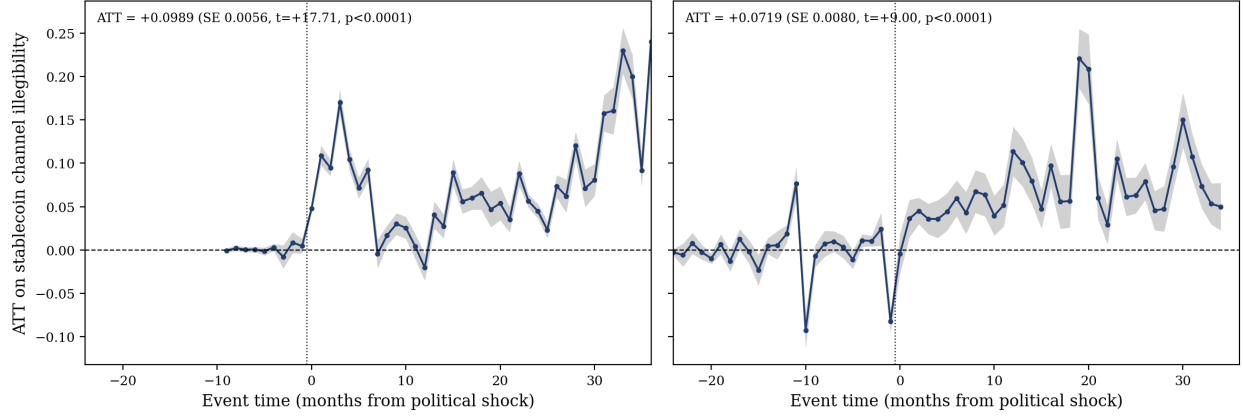


Figure 6: Dynamic event-study profiles for Hong Kong (NSL, 2020-06) and Russia (SWIFT, 2022-02). Shaded bands are 95% pointwise confidence intervals from country-clustered standard errors. The vertical dashed line marks the treatment month.

Hong Kong and Russia

The trajectories for Hong Kong and Russia are similar: pre-event coefficients are indistinguishable from zero, while post-event coefficients jump within the first month and accumulate to large positive values within two years. In raw terms, Hong Kong’s small-transaction stablecoin volume in 2019 ran 99.8% through centralized exchanges, the most state-observable channel; by 2023, that share had fallen to roughly 71% as users migrated to peer-to-peer venues and on-chain stablecoin wallets. Russia’s response to the February 2022 SWIFT exclusion is similar one year out.

Nigeria: a political shock and a regulatory shock

Nigeria supplies relatively clean identification: two financial-system shocks in one country, four months apart, of theoretically opposite type. The first was the November 2020 EndSARS aftermath, when the Central Bank of Nigeria (CBN) obtained a court order freezing protest organizers’ accounts and the State Security Service intensified financial enforcement against named dissidents; the second was the February 2021 CBN circular ordering banks to close cryptocurrency-exchange accounts. The EndSARS freeze is a political shock, signaling predatory state activity without directly restricting regulated banking. The CBN circular is a regulatory action that closes intermediated channels by restricting availability without sending any new political-cost signal. Hypothesis 1’s shock-response claim predicts a routing response to the first and a null to the second.

Reading the Nigeria EndSARS event study requires distinguishing between two periods. The EndSARS-aftermath window proper (event-time 0–24, Nov 2020 – Oct 2022) ends before an independent set of monetary-system shocks, which include the 2022–23 naira-redesign and demonetization crisis, the February 2023 Buhari–Tinubu transition, and the June 2023 FX-market liberalization, and which would affect cryptocurrency routing for reasons unrelated to the November 2020 political-cost mechanism. In that window, the EndSARS aftermath produces a positive shift in the aggregate small-transaction illegibility index but a null effect in the stablecoin-specific channel: citizens move broadly toward less-observable venues in the two years after the freezes. The full 0–49 window adds further positive shifts, but these sit beyond event-time 24 and track the post-2023 shocks rather than the political signal. Two distinct processes thus operate in the Nigerian data. First, a political-cost response to EndSARS on broad small-transaction routing; second, a separate

post-2023 shift on stablecoin routing. Our argument pertains only to the first. Figure 7 plots both, with the two windows shaded: the aggregate-index response sits inside the EndSARS window, the stablecoin-channel response in the post-2023 window.

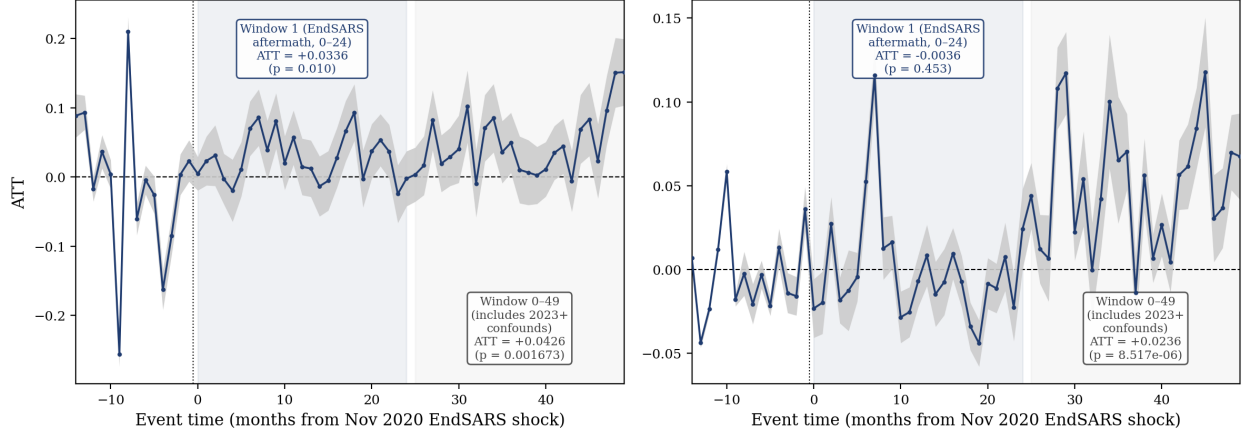


Figure 7: Dynamic event-study profiles for the Nigeria-EndSARS specification (trigger month = November 2020) on two outcomes. Left: aggregate small-transaction illegibility index (L_{agg}^{small}). Right: stablecoin channel illegibility (L^{stable}). Light-blue shading marks the EndSARS-aftermath window (event time 0–24, Nov 2020 – Oct 2022); light-orange shading marks the post-2023 confounded window (event time 25–49). Solid lines are point estimates; shaded bands are 95% pointwise confidence intervals from country-clustered standard errors. The aggregate-index response appears within the EndSARS window; the stablecoin-channel response is concentrated in the post-2023 window and is not attributable to the EndSARS shock.

Pooled cohorts and episode heterogeneity

Hong Kong, Russia, and Nigeria show that the mechanism is identifiable when the shock is unambiguous. To test whether it operates generally, we pool ERT-identified events into three cohorts (full lists in Appendix C.1): autocratization, democratization, and regime breakdown. Our framework predicts that citizens confronting autocratization and regime breakdown shift toward less observable venues as the political cost of legibility rises. The democratization cohort is a placebo: citizens leaving autocracy for democracy face a *falling* political cost and have no reason to re-route, so a null result there confirms that the autocratization findings reflect a political-cost mechanism rather than a generic news-shock response or estimation artifact.

Our theory makes a further prediction that helps shape how to interpret the pooled estimate: the autocratization response should concentrate in episodes still unfolding during our window, where citizens are actively updating their assessment of the state, and fade in episodes that have already concluded. Because global routing illegibility is rising over the period, a pooled estimate that averages ongoing and completed episodes may appear negative relative to the never-treated trend, even as treated-country illegibility rises in absolute terms. We therefore expect a negative pooled coefficient that resolves into a strong ongoing-episode effect once the cohort is split.

Table 5 reports the pooled post-period treatment effects for all three cohorts on the channel and instrument margins.

Table 5. Pooled cohort effects of political shocks on routing illegibility and instrument share.

	Routing-channel margin			Instrument margin
Cohort	$L^{\wedge}\text{stable}$	$L^{\wedge}\text{small_agg}$	Layered-evasion gap	Stablecoin share
Autocratization ($n = 14$)	negative**	negative, n.s.	negative, n.s.	not significant
Democratization ($n = 9$, placebo)	not significant	not significant	not significant	not significant
Documented regime transitions ($n = 14$)	negative, n.s.	negative*	positive*	not significant

*Callaway–Sant’Anna doubly-robust difference-in-differences with country-clustered standard errors and monthly trigger timing. Cohort and event-date lists in Appendix C. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

As predicted, the democratization placebo is null on every outcome. The autocratization cohort’s pooled effect on $L^{\wedge}\text{stable}$ is negative and significant, as expected. The regime-transitions cohort yields a positive estimate of the layered-evasion gap ($L^{\wedge}\text{stable} - L^{\wedge}\text{nonstable}$), significant at the .10 level: citizens route their stablecoin volume through less-observable channels than their non-stablecoin volume, stacking the dollar-instrument and illegible-routing margins on the same transactions. On its own, instrument margin (the share of holdings in stablecoins) is null across all three cohorts: citizens are not detectably shifting into stablecoins over these short event windows, even though the cross-section shows that pattern (Table 2).

To test the heterogeneity prediction directly, we use V-Dem’s ERT codes, which allow us to identify whether the autocratization episode we observe in our sample is ongoing (still unfolding), severe (completed), or averted. This splits our 14-country cohort into six ongoing, six severe-completed, and two averted cases (lists in Appendix C.1).

Table 6. Autocratization-cohort heterogeneity by V-Dem episode outcome.

Subset	n	Effect on $L^{\wedge}\text{stable}$
Pooled (all autocratization onsets)	14	negative**
Ongoing (outcome = ongoing)	6	negative***
Breakdown (outcome = ongoing or severe)	12	negative**
Severe (outcome = severe only)	6	negative, n.s.

*Single-cohort Callaway–Sant’Anna doubly-robust difference-in-differences for each subset. The ongoing-subset effect is more than three times as large as the severe-subset effect; the pooled effect is the cohort-weighted echo of the two subgroups. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

The pattern matches the prediction. The ongoing subset’s effect on $L^{\wedge}\text{stable}$ is statistically significant; the severe subset is similarly signed with confidence intervals that span zero. Treated-country illegibility rises over the window in raw terms, but the global control trend rises faster, yielding a negative coefficient relative to the never-treated comparison: the mechanism is most visible where citizens have room to move, and the institutional environment is visibly changing. The pooled estimate survives a random-permutation placebo (Appendix C.5); dynamic event-study profiles for the ongoing and severe subsets are in Appendix C.

The within-country evidence corroborates the cross-sectional results using a different research design. Citizens re-route in real time when domestic institutional conditions visibly change; the response is concentrated on actively unfolding episodes rather than completed ones; the mechanism appears in stablecoin routing rather than in instrument shares; and the democratization and regulatory-action placebos are null. Taken together, these findings support Hypothesis 1’s claim that democratic accountability shapes routing-channel choice both in equilibrium and during periods of transition.

VI.3 Hypothesis 2: Sanctions and Disintermediation

Sanctions act on disintermediation through two opposing forces. On the demand side, like domestic predation, sanctions raise the political cost of being observed and push citizens toward less observable venues. On the supply side, sanctions degrade the infrastructure that those venues depend on, such as correspondent banking withdrawal, exchange exits, and fiat on-ramp closures, all of which shrink the routes out. Disintermediation therefore rises as sanctions tighten, until supply-side damage overtakes the citizen response and the relationship reverses: an inverted-U in sanctions intensity, whose rising portion democratic accountability amplifies because citizens with more room to substitute respond more strongly.

Table 7 reports the inverted-U specification: a positive linear and a negative quadratic coefficient on sanctions intensity.

Table 7. Sanctions intensity has a non-linear effect on disintermediation.

Variable	Inverted-U
Lagged GSDB intensity	+0.0306*** (0.0082)
Lagged GSDB intensity squared	−0.0071*** (0.0017)
Clean Elections Index	−0.0778* (0.0342)
N	4,242

*Inverted-U specification on deep disintermediation (all transactions). Same controls as Table 1 (full model in Appendix E.3). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

The estimated peak occurs at a log-intensity of roughly 2.14, corresponding to about seven to eight sanctioning countries. The peak cleanly separates the distribution of sanctions: Most sanctioned countries face only a small number of senders: 65 percent of sanctioned country-months involve one to three sanctioning states, all on the rising side of the curve. A smaller set of targets is subject to comprehensive multilateral regimes²¹: Russia (over forty senders, rising to forty-six after the 2022 invasion), Iran (three through 2021, then more than thirty after 2022), Belarus (one in 2019, over forty by 2023), and Myanmar (four before the 2021 coup, thirty-two after). These heavily sanctioned targets, representing about one-fifth of the sample, lie on the descending portion of the

²¹The sender count comes from the Global Sanctions Database, which records multilateral sanctions (UN, EU) as a separate sender–target dyad for each member state. At the high end the count therefore reflects coalition and bloc membership rather than independent national decisions, reaching roughly 190 for targets under UN financial sanctions. The inverted-U is estimated on log-intensity, which compresses this upper tail; the peak at seven to eight senders falls in the small-coalition range, where the count is unaffected by bloc aggregation.

curve, where supply-side damage dominates. Few targets fall between these two groups, so the peak is best interpreted as the transition from small-coalition to comprehensive sanctions, rather than as a level at which many countries are observed. Figure 8 plots the curve with a 95% confidence-interval band.

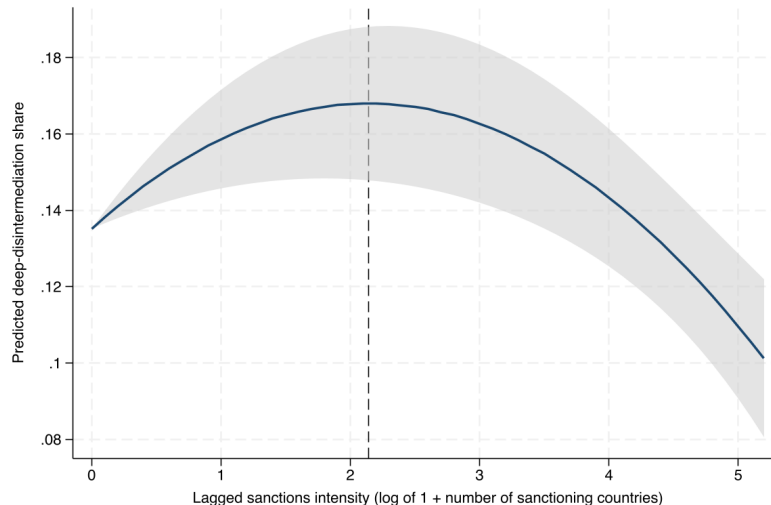


Figure 8: Sanctions and disintermediation: the inverted-U. Predicted contribution to deep disintermediation from Table 7 Column 2, with 95% confidence interval band computed by the delta method. Vertical dashed line marks the estimated peak at log-intensity 2.14 (~7 to 8 sanctioning countries).

The inverted-U so far treats every regime alike, but the framework makes a sharper prediction about who responds on its rising portion. When sanctions raise the cost of staying legible, citizens with greater adaptive capacity (a more open internet, a state less able to suppress the exit) can substitute toward disintermediated channels more readily, so the same increase in sanctions moves them further. Democracies should therefore amplify the demand-side response rather than dampen it (Implication 2b). Table 8 tests this by including the interaction between the Clean Elections Index and lagged sanctions intensity. The interaction coefficient is positive but imprecisely estimated (+0.036, $p = 0.13$), so the cross-partial on its own does not establish amplification; the evidence comes instead from the marginal-effect profile it implies.

Table 8. Democracy and the demand-side response to sanctions.

Variable	Inverted-U	Dem $\times$ Sanctions
Lagged GSDB intensity	+0.0306*** (0.0082)	+0.0114 (0.0160)
Lagged GSDB intensity squared	−0.0071*** (0.0017)	−0.0037 (0.0029)
Clean Elections Index	−0.0778* (0.0342)	−0.0944** (0.0398)
Lagged GSDB $\times$ Democracy	—	+0.0356 (0.0236)
Lagged GSDB squared $\times$ Democracy	—	−0.0065 (0.0051)

Variable	Inverted-U	Dem $\times$ Sanctions
N	4,242	4,242

Same specifications and controls as Table 7 (full models in Appendix E.3).

The amplification is clearest in the implied marginal effects. Figure 9 traces the marginal effect of a sanctions increase across the democracy distribution, evaluated at a moderate sanctions intensity. It rises monotonically — from about +0.006 in the most autocratic countries to about +0.033 in the most democratic — and clears zero (at 90 percent confidence) for moderately democratic countries and above, while remaining indistinguishable from zero at the autocratic end. This is the expectation of Implication 2b: democratic citizens respond more strongly to a marginal sanctions increase, through the adaptive-capacity channel $\psi(\delta)$ of Appendix B.

Two identification concerns arise. The first is reverse causality: countries may be sanctioned partly in response to crypto-mediated evasion, which could by itself generate a positive correlation between sanctions intensity and disintermediation. The graded marginal-effect profile is hard to reconcile with this explanation. Reverse causality would inflate the overall correlation, but it offers no reason why the response should rise with democratic accountability, and that profile reflects citizen demand rather than the government behavior that triggers sanctions in the first place.

The second concern is that the inverted-U might indicate a general decline in a target’s external relations rather than financial exclusion specifically. Two robustness checks are provided in Appendix D.3 to address this issue. First, restricting sanctions intensity to Western senders alone (the United States, EU member states, the United Kingdom, Canada, Australia, and Japan) preserves the inverted-U at a comparable magnitude. Second, a placebo constructed from non-financial sanctions yields null coefficients on both terms, as it should if the operative mechanism is financial exclusion rather than general diplomatic strain.

VI.4 Combined Model

We perform two final checks to confirm that the reported effects of democracy and sanctions are distinct and not attributable to an omitted confounder. The first jointly estimates the H1 baseline and the H2 inverted-U. If the two were a single effect entering through different regressors, including both would make one absorb the other; instead, in Table 9, the democracy coefficient remains negative and significant alongside the sanctions inverted-U. The two hypotheses identify distinct margins of the democracy effect, not the same one twice.

Table 9. Combined model.

Variable	H1 Baseline	+ Sanctions
Clean Elections Index	−0.0630* (0.0366)	−0.0778* (0.0342)
Lagged GSDB intensity	—	+0.0306*** (0.0082)
Lagged GSDB intensity squared	—	−0.0071*** (0.0017)
N	4,242	4,242

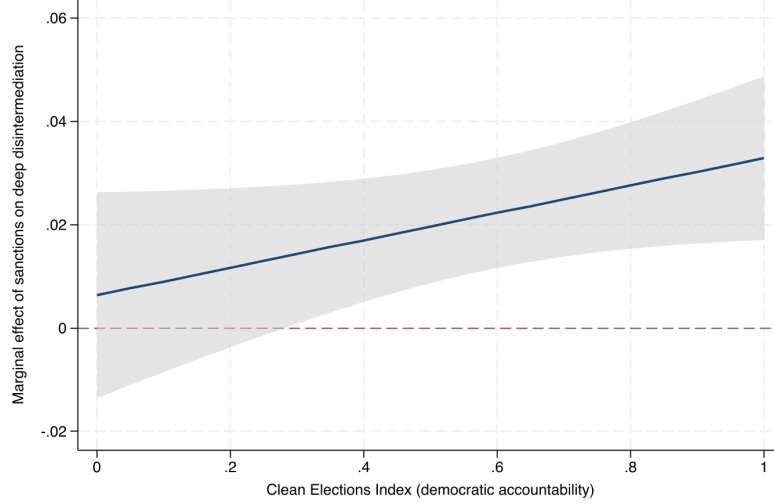


Figure 9: Marginal effect of sanctions on deep disintermediation, by democracy. Marginal effects implied by the Table 8 interaction specification, evaluated at the 25th percentile of nonzero sanctions intensity and traced across the Clean Elections Index. The shaded band is the 90% confidence interval; the dashed line marks zero.

*Same specification and controls as Table 1 (Clean Elections Index and sanctions terms shown; full models in Appendix E.1 for the H1 baseline and Appendix E.3 for sanctions). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

The second check confirms that the democracy effect is not an artifact of a single outcome construction: it appears across both channels identified by the formal model. Table 10 shows a negative, significant Clean Elections Index coefficient on the instrument channel (the USDC share) and on the exchange channel (deep disintermediation in small transactions), the two channels moving together as expected.

Table 10. The democracy effect on both margins.

Variable	USDC Share	Deep Disint (Small)
Clean Elections Index	−0.0528* (0.0317)	−0.1007** (0.0398)
N	4,242	4,242

*Same specification and controls as Table 1 (full models in Appendix E.2). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

The quantitative results document the citizen side of the relationship: where accountability is weak, citizens route a larger share of activity through channels the state cannot read. The same institutional weakness should shape the state’s behavior, too. A government facing few constraints both provokes more exit and has the latitude to extend its legibility infrastructure when an opening arises. The next section turns to that state-side response, tracing how low-accountability governments use episodes of monetary stress to steer cryptocurrency activity into channels they can monitor.

VII. Case Evidence on the Regulatory Crackdown Mechanism

If low-accountability states extend surveillance under fiscal stress, the tendency should be visible cross-sectionally: cryptocurrency banking bans should be more common in autocracies. They are. Among non-advanced economies, autocracies — countries whose mean Clean Elections Index falls below 0.5 — are about twice as likely to have ever imposed a crypto banking ban as democracies: 12.2 percent versus 5.6 percent (Table 11). We read this as an autocracy regularity.

Table 11. Cryptocurrency banking bans are more common in autocracies.

Country group	Ever imposed a banking ban	Countries
Democracies (CEI mean ≥ 0.5)	5.6%	54
Autocracies (CEI mean < 0.5)	12.2%	74

Non-advanced economies, country level. Autocracy = country-mean Clean Elections Index below 0.5. Descriptive shares of countries that have ever imposed a crypto banking ban; no significance test.

This cross-country gap is consistent with autocracies’ greater willingness to extend financial surveillance, but it is descriptive and says nothing about timing or mechanism. The three cases that follow, Nigeria 2021, Türkiye 2021, and Argentina 2022, supply what the cross-section cannot: each combines acute monetary stress with a state response that extends financial surveillance through new regulation of cryptocurrency intermediaries, and a documented effect on citizen channel choice in the months that follow. These cases provide mechanism evidence, showing that the regulatory-crackdown story corresponds to identifiable state behavior.

Nigeria 2021

The macroeconomic backdrop in Nigeria was a managed exchange-rate regime under acute strain. The Central Bank of Nigeria’s (CBN) official naira–U.S. dollar rate, held near ₦380 through most of 2020, was devalued to roughly ₦410 in late 2020, while the parallel-market rate widened to ₦480–₦500 by January 2021 as oil receipts collapsed and dollar liquidity dried up at the I&E window. Against this background, the CBN issued Letter Ref. BSD/DIR/PUB/LAB/014/001, dated 5 February 2021, to all Deposit Money Banks, Non-Bank Financial Institutions, and Other Financial Institutions, reminding them that “dealing in cryptocurrencies or facilitating payments for cryptocurrency exchanges is prohibited” and directing them to “identify persons and/or entities transacting in or operating cryptocurrency exchanges within their systems” and to “ensure that such accounts are closed immediately” (Central Bank of Nigeria 2021). Enforcement began immediately: within weeks, named exchanges including Binance, Bundle, and Quidax reported the closure of correspondent banking relationships with Nigerian institutions.

The citizen response was rapid and channel-substituting rather than activity-reducing. Chainalysis’s 2021 *Geography of Cryptocurrency* report ranked Nigeria sixth globally on its adoption index and documented an outsized share of activity routed through peer-to-peer marketplaces, with volumes on Paxful and LocalBitcoins continuing to rise through Q2 2021 even as on-ramp access at licensed banks collapsed (Chainalysis 2021). The institutional context was permissive: during the Buhari administration, the CBN, under Governor Godwin Emefiele, repeatedly served as an instrument of administrative policy rather than an independent monetary authority. In October–November 2020, the CBN had obtained a court order freezing the accounts of #EndSARS protest

organizers (Amnesty International 2021), an episode consistent with a broader pattern of executive-driven instrumentalization of regulatory authority (Lewis 2007; Page 2021). These two events — the November 2020 account-freeze episode and the February 2021 crypto-banking ban — are the within-Nigeria comparison analyzed above: the political-cost shock produces a measurable routing response while the regulatory ban produces none, the divergence the framework predicts.

Türkiye 2021

The Turkish lira entered 2021 weak, down from roughly TL5.95/USD in early 2020 to TL7.40 by year-end, but the proximate trigger was political. On 20 March 2021, President Erdoğan dismissed the Governor of the Central Bank of the Republic of Turkey (CBRT), Naci Ağbal, by decree two days after he raised the policy rate by 200 basis points, causing the lira to drop by roughly 15 percent (Reuters 2021; CNN Business 2021). Twenty-six days later, the CBRT published a regulation (*Resmî Gazete* No. 31456, 16 April 2021, effective 30 April) prohibiting the use of crypto-assets for payments and barring payment and electronic-money institutions from facilitating such use (Central Bank of the Republic of Türkiye 2021). Citizens responded before it took effect: Chainalysis recorded Turkish trading volumes climbing through March and April, and the 22 April collapse of the domestic exchange Thodex — whose founder fled with an estimated \$2 billion in client assets — pushed migration toward foreign exchanges and stablecoin self-custody rather than back into the banking channel (Chainalysis 2021; Reuters 2021). Ağbal’s removal was the third governor dismissal in twenty months, under a 2018 decree granting the president direct authority to appoint and dismiss governors. Each removal was followed by depreciation and tighter administrative controls (Acemoglu and Ucer 2015; Esen and Gumuscu 2016).

Argentina 2022

Argentine inflation accelerated from roughly 50 percent year-on-year at the start of 2022 to 71 percent by July and approximately 95 percent by December, while the official peso/USD rate moved from ARS103 in January to ARS177 by year-end and the parallel “dólar blue” widened from ARS208 to nearly ARS350 over the same period, all within a tightening *cepo cambiario* exchange-control regime restricting retail dollar purchases to USD 200 per month (Reuters 2022). On 2 May 2022, two large private banks (Banco Galicia and Brubank) announced retail crypto-trading services; three business days later, on 5 May 2022, the Banco Central de la República Argentina issued *Comunicación “A” 7506*, “Servicios complementarios de la actividad financiera y actividades permitidas — operaciones con activos digitales,” prohibiting financial entities from carrying out or facilitating the execution of operations with digital assets, including cryptoassets and those whose yields are determined based on the variations they register.” (Banco Central de la República Argentina 2022). *Comunicación “A” 7507* followed days later, restricting collective investment vehicles in the same direction.

The citizen response was an increasing move to disintermediated exchanges. Chainalysis ranked Argentina among the top fifteen countries on its 2022 adoption index, with a disproportionate share of activity in dollar-pegged stablecoins (USDT and USDC) functioning as informal savings instruments, and with P2P platform volumes (Binance P2P and the domestic wallet Lemon in particular) rising sharply through Q3 2022 (Chainalysis 2022). The institutional context was an unusually weak BCRA: although de jure independent, its directorate served at the pleasure of the executive and operated under pressure to monetize fiscal deficits, a pattern that intensified after Economy Minister Martín Guzmán resigned in early July 2022 and Sergio Massa assumed an enlarged Ministry of Economy portfolio in August 2022 with direct authority over exchange-

control policy (Gerchunoff and Llach 2018; Edwards 2010). The speed of the A 7506 response, three business days after two private banks launched retail crypto trading, is consistent with our theory: a monetary authority moving to close a substitution channel the moment it opened.

The three cases illustrate a common mechanism. Monetary stress raises the salience of cryptocurrency markets as instruments of capital flight and store-of-value protection, and states facing few institutional constraints respond by extending financial surveillance to these channels through banking bans, payment restrictions, or licensing requirements that route activity through state-legible venues. The citizen response depends on the form of the intervention: regulatory measures that close intermediated channels push activity into disintermediated ones through availability, while political-cost shocks that signal predatory state intent pull citizens toward them through choice.

VIII. Conclusion

We begin with a puzzle. From Nigeria to Hong Kong to Russia, citizens route cryptocurrency through venues with wider spreads, lower liquidity, and no consumer protection, willingly paying more for the alternative. This pattern cannot be explained by standard models of disintermediation, which are built on efficiency arbitrage. The standard account of financial disintermediation views it as a form of efficiency arbitrage: yield differentials drive substitution toward a cheaper option (Diamond 1984). Cryptocurrency disintermediation is puzzling as it operates in the opposite direction: citizens who route transactions through decentralized exchanges accept higher friction, lower liquidity, and the absence of consumer protection. The puzzle dissolves once we recognize that intermediation generates a politically valuable byproduct — legibility — whose political cost varies with institutional constraints on state power. Democratic accountability constrains the executive use of that legibility on two margins at once. It lowers the equilibrium political cost of being observed, reducing disintermediation under normal conditions (Hypothesis 1). And it moderates citizens’ responses to international financial sanctions, dampening the demand for illegibility where institutional constraints provide reassurance against escalation (Implication 3b).

This argument extends the monetary-politics literature on credible commitment (Cukierman, Webb, and Neyapti 1992; Maxfield 1997; Broz 2002; Bodea and Hicks 2015), which has focused on the conditions under which states commit not to inflate. We identify a parallel commitment problem: the conditions under which states commit not to weaponize the financial system against its users. The cryptocurrency market makes both commitment failures simultaneously observable. Citizens substitute toward dollar-denominated instruments when the value commitment is not credible, and toward illegible channels when the access commitment is not credible. That the two responses are empirically separable (operating on different dimensions of the same transaction, responding to different institutional variables, and concentrating in different market segments) is what distinguishes the cryptocurrency setting from earlier episodes of capital flight, dollarization, or informal finance.

Our argument also bears on policy debates about cryptocurrency regulation. The standard framing treats cryptocurrency primarily as a tax-evasion, money-laundering, and consumer-protection problem, to be addressed through KYC requirements, exchange licensing, and reporting mandates that incorporate activity into the existing legibility architecture. That framing is incomplete. In democratic settings, regulatory integration is defensible: the political cost of legibility is low, and the gains in consumer protection and tax administration plausibly exceed the costs to financial privacy. In low-accountability environments, however, the same architecture functions as an instru-

ment of political control, and citizens who route activity through disintermediated channels are exiting an architecture that has demonstrably been turned against them. The degree of democratic accountability determines whether regulatory integration ultimately serves the citizens it ostensibly protects.

Two extensions follow naturally. Our framework treats domestic political institutions as exogenous to cryptocurrency adoption, but at sufficient scale, the access-commitment problem may in turn condition institutional outcomes: when citizens can credibly threaten exit through illegible channels, the political return to surveillance infrastructure falls, and the incentive to constrain executive power may strengthen. Our framework also stops short of modeling the strategic state behavior illustrated in Section VII: treating regulatory-crackdown intensity as a choice conditioned on accountability would sharpen predictions about which states crack down, when, and how.

States and markets are the stuff of political economy. We know that citizens vote with their wallets — selectively, partially, and reversibly — on whether the institutions that govern their money can be trusted not to turn it against them. Where that trust is absent, citizens are building the exit in plain sight.

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Appendix A: Blockchain Technology Primer

This appendix provides technical background: what a blockchain is, how proof-of-work and proof-of-stake consensus work, what custody means in this setting, why the Tier 0 through Tier 3 classification of footnote ²² tracks an underlying technical distinction rather than a regulatory label, and what separates the two instrument types, stablecoins and non-stablecoins, that the paper treats as a second margin.

A.1 The double-spending problem

A physical banknote can only be handed to one person at a time. A digital token is a string of bits, and bits can be copied. Without a mechanism to prevent that copying, the recipient of a digital payment cannot know whether the sender has already spent the same funds elsewhere. From the 1980s through 2008, every working digital payment system solved this problem the same way: a single trusted institution maintained an authoritative ledger and refused to honor any transfer whose source had already been spent (Chaum 1983; DigiCash; e-gold). The institution was always something a state could compel. Digital money was therefore inherently legible to state authority — not by policy choice but as a technical entailment of the only working solution to double-spending.

A.2 Hashes, blocks, and the chain

Nakamoto’s (2008) contribution dissolved that entailment. The key technical building block is the cryptographic hash function — a procedure that takes any input and produces a short, fixed-length fingerprint with two properties: identical inputs always produce the same fingerprint, and finding two distinct inputs that share a fingerprint is computationally infeasible. A “block” is a batch of recent transactions together with the fingerprint of the immediately preceding block. The fingerprint of the new block, therefore, depends on the fingerprint of every previous block, all the way back to the first (“genesis”) block. The chain is this sequence.

The property loosely called “immutability” falls out directly. Altering any historical transaction changes that block’s contents, which changes its fingerprint, which breaks the fingerprint reference in the next block, and the next, and the next. The only way to alter a past entry is to recompute every subsequent block faster than the rest of the network extends the current chain — not a feasible operation for any single actor on a sufficiently large network.

A.3 Consensus without an authority

The remaining technical problem is which set of new transactions the network agrees to record next, in the absence of any central authority. Two consensus protocols are used in practice: proof-of-work (Bitcoin since 2009) and proof-of-stake (Ethereum since September 2022). Under proof-of-work, computers on the network compete to find a numerical value that, when combined with

²²The legibility weight τ_k is assigned to each Chainalysis service category k based on the strength of the identity link the venue transmits to the state authority. The mapping is as follows. Tier 0 ($\tau = 0.0$): centralized exchange, hosted wallet, merchant services. Tier 1 ($\tau = 0.2$): lending, lending contract. Tier 2 ($\tau = 0.4$ – 0.5): smart contract, token smart contract, ICO, NFT platform collection (each 0.4); no-KYC exchange (0.5). Tier 3 ($\tau = 0.6$ – 0.7): decentralized exchange, decentralized exchange contract, bridge (each 0.6); peer-to-peer exchange, ATM (each 0.7). Tier 4 ($\tau = 0.9$ – 1.0): mixing, protocol privacy (each 0.9); darknet market, sanctioned entity, sanctioned jurisdiction, sanctions, illicit actor/organization, fraud shop, scam, child abuse material (each 1.0). The mining pool, infrastructure-as-a-service, unclassifiable “other” volume, and the gambling service category are excluded from gradient construction. The gambling exclusion follows from per-category diagnostics reported in Section VI.1.

a candidate block, produces a fingerprint below a target threshold. The search has no shortcut: the first machine to succeed broadcasts the block; the rest of the network verifies it, and the chain extends by one. Under proof-of-stake, validators are selected to propose the next block in proportion to the amount of native tokens they commit as collateral, and misbehavior costs collateral.

Both protocols deliver the same political property. The network reaches consensus through a rule no participant controls: no headquarters to subpoena, no CEO to compel, no addressable entity to whom a freeze order can be served. Court orders to “freeze” cryptocurrency are served on the intermediaries that sit on top of the protocol, not on the protocol itself.

A.4 Wallets: custodial versus non-custodial

A wallet is not a balance of tokens. It is a pair of cryptographic keys — one public, one private — that controls a single address on the blockchain. The address is derived from the public key; the private key authorizes transfers out of the address. Whoever holds the private key controls the funds, and the disintermediation variable measures who that is.

In a *custodial* wallet, an intermediary — typically a centralized exchange — holds the private key on the user’s behalf. The user logs in with a username and password; the exchange signs transactions when instructed. A state that can compel the exchange can therefore freeze the address, disclose its contents, or seize its balance, regardless of any cooperation from the customer. In a *non-custodial* wallet, the user holds the private key directly — typically as a string of words memorized or written down, or in a hardware device that signs transactions offline. A state that wants to interfere with these funds has no intermediary to compel; it can prosecute the user for refusing to disclose the key (jurisdictions vary on whether this is permissible) but it cannot reach the address through the architecture itself.

Tier 0 services in footnote ²³ are services in which an intermediary holds the keys (centralized exchange, hosted wallet, merchant services). Tier 3 services — decentralized exchanges, peer-to-peer platforms, and cryptocurrency ATMs — assume the user has already moved funds into a non-custodial wallet that the venue does not control. Figure 10 illustrates the two pathways side by side.

A.5 What the ledger records, and what it does not

Every transfer on the Bitcoin and Ethereum chains is recorded: sending address, receiving address, amount, timestamp, transaction fee. The full record is publicly downloadable; any laptop with sufficient storage can replay the entire history from January 2009. There is no equivalent in traditional finance, where transaction history is held privately by the customer’s bank.

What the ledger does not record is the legal person associated with each address. The on-chain entry is `1A1zP1eP5QGefi2DMPTfTL5SLmv7Divf`, not “Jane Doe.” The link between an address and a name

²³The legibility weight τ_k is assigned to each Chainalysis service category k based on the strength of the identity link the venue transmits to the state authority. The mapping is as follows. Tier 0 ($\tau = 0.0$): centralized exchange, hosted wallet, merchant services. Tier 1 ($\tau = 0.2$): lending, lending contract. Tier 2 ($\tau = 0.4$ – 0.5): smart contract, token smart contract, ICO, NFT platform collection (each 0.4); no-KYC exchange (0.5). Tier 3 ($\tau = 0.6$ – 0.7): decentralized exchange, decentralized exchange contract, bridge (each 0.6); peer-to-peer exchange, ATM (each 0.7). Tier 4 ($\tau = 0.9$ – 1.0): mixing, protocol privacy (each 0.9); darknet market, sanctioned entity, sanctioned jurisdiction, sanctions, illicit actor/organization, fraud shop, scam, child abuse material (each 1.0). The mining pool, infrastructure-as-a-service, unclassifiable “other” volume, and the gambling service category are excluded from gradient construction. The gambling exclusion follows from per-category diagnostics reported in Section VI.1.

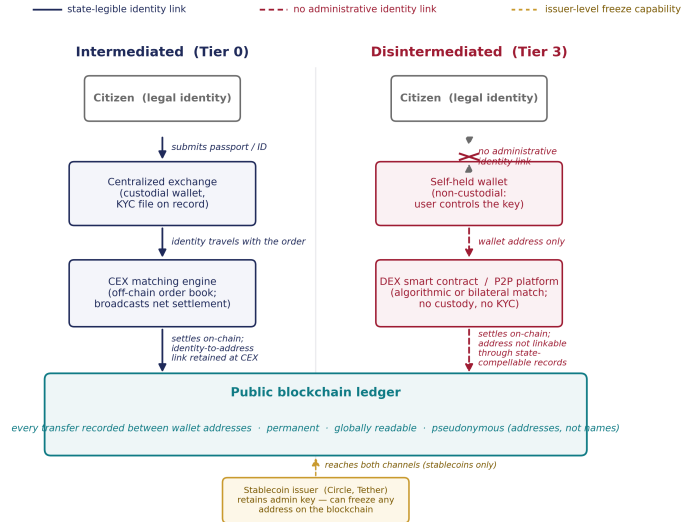


Figure 10: Two parallel pathways for the same cryptocurrency purchase. *Left:* the citizen submits identity documents to a centralized exchange, which custodies the wallet, matches orders against its internal book, and settles the net position on the public blockchain. The identity-to-address link is retained on the exchange’s servers and is compellable by any authority with jurisdiction over the exchange. *Right:* the citizen acquires funds through a peer-to-peer or decentralized exchange venue that performs no KYC, into a wallet whose private key she holds herself. The on-chain entry exists, but no administrative record links it to her legal identity. The blockchain layer is shared across both pathways. The dashed gold arrow indicates the issuer-layer freeze capability that reaches stablecoin balances regardless of channel.

is established outside the protocol: by the KYC file at a regulated exchange, by an IP address logged when a transaction is broadcast, or by a counterparty whose identity is already known. This is what is meant by the standard description of cryptocurrency as pseudonymous rather than anonymous. The chain-analysis industry — Chainalysis, Elliptic, CipherTrace, discussed in footnote ²⁴ of the main text — exists because the on-chain record is so transparent; its business is not to reveal hidden data but to assemble external information that connects public addresses to the legal persons who control them.

No cryptocurrency channel is anonymous at the ledger level. What varies across channels is whether a state can compel disclosure of the identity link through a regulated intermediary. Where the user has acquired and moved funds entirely through Tier 0 venues, the link sits on a server at the exchange and can be subpoenaed; where funds have been acquired through a Tier 3 venue and never touched a regulated exchange, no such administrative record exists.

A.6 Mapping technology to legitimacy tiers

Three venue types — CEX, DEX, and P2P — describe nearly all the activity examined in the paper.

A *centralized exchange* (CEX) operates like a bank for cryptocurrency: it accepts customer fiat

²⁴Forensic firms such as Chainalysis, Elliptic, and CipherTrace (contracted by the U.S. Treasury, DEA, FBI, and numerous foreign governments) have built substantial businesses on the underlying transparency of the blockchain. Their core technique is clustering: wallet addresses are grouped by transaction patterns, and each cluster is identified through external information.

deposits, custodies customer crypto holdings in exchange-controlled wallets, maintains an off-chain order book that matches orders internally, and broadcasts only net settlement to the blockchain. KYC is a regulatory requirement in essentially every advanced economy and in most emerging markets where exchanges operate under license. From the state’s perspective, a CEX is functionally identical to a bank: it knows who its customers are, what they hold, what they have transacted, and is required to report on demand. Centralized exchanges, hosted wallets, and merchant-service providers constitute Tier 0.

A *decentralized exchange* (DEX) is a smart contract — software running on a blockchain — that swaps one token for another against a pool of tokens supplied by other users. The contract has no operator and no balance sheet; it executes deterministically against transactions sent to it from non-custodial wallets. There is no account, no log-in, no identity collection. A *peer-to-peer* (P2P) platform matches buyers and sellers for direct bilateral settlement; payment is made outside the platform (bank transfer, mobile money, cash), and the platform’s role is matchmaking and, in some cases, escrow. KYC is minimal, often a phone number, and frequently absent. DEX contracts, their cross-chain bridges, and P2P platforms together constitute Tier 3.

One vulnerability does not respect the channel distinction. The major dollar-pegged stablecoins — USDC and USDT — are issued by private companies that retain administrative keys allowing them to freeze any address holding their token on-chain, regardless of whether the address belongs to a CEX customer or a self-custody holder. Circle exercised this capability after the August 2022 Treasury designation of Tornado Cash, freezing every USDC address associated with the sanctioned smart contract within hours. This issuer-layer vulnerability operates orthogonally to the intermediated/disintermediated distinction; it is why the disintermediated-stablecoin cell of the four-cell decomposition in the main text is empirically thin.

Returning to Figure 10: the same purchase, executed through either pathway, produces an entry on the same blockchain. What differs is whether an administrative record links that entry to the citizen. The tier classification — Tier 0 for CEX-mediated activity, Tier 3 for DEX/P2P activity — tracks this single distinction.

Appendix B: The Formal Model

Section III states the citizen’s problem, the interior optimum, the inverted-U emergence condition, and the amplification result. This appendix derives them in full: the first-order condition, the dominance condition under which Hypothesis 1 holds, the derivation of the inverted-U and its peak, the full demand-region and descending-portion derivations of the amplification result, and the multi-tier and transaction-size extensions that generate Implications 1a, 1c, and 2a. Each derivation includes the sufficient conditions under which the relevant prediction holds.

B.1 Setup

Consider a representative citizen in country i at time t executing a cryptocurrency transaction of value V . The citizen allocates a share $\alpha \in [0, 1]$ of the transaction to intermediated channels and the residual share $(1 - \alpha)$ to disintermediated channels.

The intermediated channel provides a non-price benefit $B > 0$ — consumer protection, deposit

insurance, legal recourse, ease of use — available only through regulated venues. The disintermediated channel provides no such benefit and transmits no identity record to state authority.

We normalize the marginal economic cost of the intermediated channel to zero. The disintermediated channel imposes an economic cost that we model as a convex function of the disintermediated share:

$$C_D(1 - \alpha) = \frac{c_D^0}{2\psi(\delta)}(1 - \alpha)^2$$

The convexity is essential to the model’s empirical content. A linear specification yields only corner solutions ($\alpha^* \in \{0, 1\}$), which would predict that every country either fully intermediates or fully disintermediates. The convex specification admits two interpretations. At the individual level, the quadratic captures the empirical observation that deeper disintermediation requires progressively more sophisticated tooling and exposure to more illiquid markets. At the country-aggregate level, the quadratic captures the envelope across heterogeneous individual cost functions: the population-level cost of moving a marginal additional share to disintermediated channels.

The parameter $c_D^0 > 0$ captures the level of the disintermediated cost. The parameter $\psi(\delta) > 0$ is the adaptive-capacity term: in countries with greater democratic accountability, the effective cost of disintermediation is lower because citizens face fewer technical, informational, and institutional barriers to exit. We assume $\psi'(\delta) > 0$ and $\psi(\delta) > 0$ on $\delta \in [0, 1]$.

B.2 The Political Cost of Legibility

The political cost of routing a transaction through the intermediated channel depends on two sources of risk:

$$\Phi(\delta, r, S) = \pi(\delta) \cdot L(r) + \eta \cdot S$$

The first term captures domestic risk. $\pi(\delta) \in [0, 1]$ is the probability that the domestic state takes predatory action against the citizen conditional on observing her financial activity. $L(r) > 0$ is the loss she suffers if predatory action occurs, where r indexes financial repression intensity. We assume $\pi'(\delta) < 0$ and $L'(r) > 0$.

The second term captures external risk. $S \geq 0$ indexes the intensity of international financial sanctions against country i , and $\eta > 0$ scales the citizen’s expected loss from a sanctioning government acting against her through the intermediated system. The scale parameter η plausibly varies with the depth of the target country’s integration into the global financial system; we treat it as a country-specific constant in the static model and revisit it in the empirical specifications, where country fixed effects absorb the cross-sectional variation.

B.3 The Citizen’s Problem and the Interior Optimum

The citizen chooses α to maximize expected utility:

$$\max_{\alpha \in [0, 1]} U(\alpha) = V - \frac{c_D^0}{2\psi(\delta)}(1 - \alpha)^2 - \alpha \cdot \Phi(\delta, r, S) + \alpha \cdot B$$

The first-order condition is:

$$\frac{\partial U}{\partial \alpha} = \frac{c_D^0}{\psi(\delta)}(1 - \alpha) - \Phi + B = 0$$

Solving for the interior optimum:

$$1 - \alpha^* = \frac{\psi(\delta) \cdot (\Phi - B)}{c_D^0}$$

The interior solution applies when $B < \Phi < B + c_D^0/\psi(\delta)$.

B.4 Result 1: The Baseline Democracy Effect (Hypothesis 1)

Setting $S = 0$ and differentiating with respect to δ :

$$\left. \frac{\partial(1 - \alpha^*)}{\partial \delta} \right|_{S=0} = \frac{\psi'(\delta) \cdot [\pi(\delta)L(r) - B]}{c_D^0} + \frac{\psi(\delta) \cdot \pi'(\delta) \cdot L(r)}{c_D^0}$$

The capacity term is positive (democratic citizens find disintermediation cheaper, raising the share they substitute into it); the baseline-cost term is negative (democratic citizens face lower political cost of legibility, reducing the share they substitute). The net effect is negative under a sufficient-condition assumption rather than as a theorem:

Assumption 1 (baseline-cost dominance). At the empirical means of the sample, the baseline-cost gradient dominates the capacity gradient:

$$\psi(\delta) \cdot |\pi'(\delta)| \cdot L(r) > \psi'(\delta) \cdot [\pi(\delta)L(r) - B]$$

Two considerations make the assumption plausible. First, the interior-solution requirement $B < \Phi$ implies that $\pi L > B$, but the gap $(\pi L - B)$ is bounded above by the maximum disintermediated share, which is far below one for almost all countries in our panel, so the baseline-cost term multiplies a moderate quantity. Second, the baseline-cost effect operates on the *level* of the political cost while the capacity effect operates on its *gradient*. The capacity-controls test in Section VI is what confirms the assumption holds in the data: the democracy coefficient survives the capacity battery, which it would not if the capacity gradient dominated.

A numerical illustration may help. Suppose $\pi(\delta) = 0.5 - 0.3\delta$ (the predation probability falls from 0.5 in a full autocracy to 0.2 in a full democracy), $\psi(\delta) = 1 + 0.5\delta$, $L(r) = 1$, $B = 0.1$, and $c_D^0 = 1$. Then $\psi'(\delta) = 0.5$, $\pi'(\delta) = -0.3$, and at $\delta = 0.5$ the dominance condition becomes $1.25 \cdot 0.3 = 0.375 > 0.5 \cdot (0.35 - 0.1) = 0.125$, comfortably satisfied. The dominance condition fails only in parameter regions where the baseline political cost is implausibly low (πL close to B , generating a degenerate interior solution) or where the capacity gradient is implausibly steep ($\psi'(\delta)$ much larger than $|\pi'(\delta)|$, which is empirically not observed).

B.5 Result 2: The Sanctions Inverted-U (Hypothesis 2)

Sanctions affect the citizen's problem through demand ($\eta \cdot S$ in Φ) and supply ($c_D^0(S) = c_D^0 + \kappa S^2$) channels. The quadratic supply-side specification deserves justification. Correspondent banks faced with a sanctioned target country make a sequential withdrawal decision: at low sanctions intensity, the compliance burden is manageable and relationships are retained; at moderate sanctions intensity, banks engage in selective de-risking that drops some relationships but retains others; at high sanctions intensity, banks exit the target jurisdiction entirely to avoid compounding regulatory liability. The transition from selective de-risking to wholesale exit is the empirical content of the quadratic term: the marginal cost of an additional sanctioning country grows faster than linearly because each marginal sanction tips additional banks past the wholesale-exit threshold. The quadratic functional form is the simplest specification consistent with this sequential structure; we treat it as an approximation rather than a deep claim about the underlying micro-structure.

The disintermediated share is:

$$1 - \alpha^*(S) = \frac{\psi(\delta) \cdot [\pi(\delta)L(r) + \eta S - B]}{c_{D,0}^0 + \kappa S^2}$$

Differentiating with respect to S :

$$\frac{d(1 - \alpha^*)}{dS} = \frac{\psi(\delta) \cdot \eta}{c_D^0(S)} - \frac{\psi(\delta) \cdot [\Phi - B] \cdot 2\kappa S}{[c_D^0(S)]^2}$$

The inverted-U emerges when the supply channel overtakes the demand channel at some interior $S^* > 0$:

$$S^* = \frac{\eta \cdot c_D^0(S^*)}{2\kappa \cdot [\Phi - B]}$$

This admits an interior solution when κ is sufficiently large relative to η . Section VI confirms the inversion at log-intensity 2.14 (approximately 7 to 8 sanctioning countries).

B.6 Result 3: Democracy Amplification on the Ascending Portion (Implication 2b)

On the ascending portion of the inverted-U the demand channel dominates, so the second (supply-side) term of the B.5 derivative is negligible and the marginal effect reduces to its first term, $\partial(1 - \alpha^*)/\partial S \approx \psi(\delta) \eta / c_D^0$. Differentiating this with respect to δ — common factor η / c_D^0 — gives the amplification result:

$$\left. \frac{\partial^2(1 - \alpha^*)}{\partial S \partial \delta} \right|_{\text{demand region}} = \frac{\eta}{c_D^0} \psi'(\delta) > 0$$

Democratic accountability amplifies the demand-side response to sanctions through the capacity channel $\psi(\delta)$. On the descending portion, infrastructure destruction is the binding constraint and the cross-partial vanishes. Democracy operates with opposite signs on the baseline and the adaptive-capacity margins; the positive cross-partial in the demand region is the empirical fingerprint of the second margin.

B.7 Multi-Tier Extension (Implication 1a)

Cryptocurrency channels are indexed by tier $k \in \{0, 1, 2, 3, 4\}$. Tiers 0 through 3 differ in economic cost, with c^k_D strictly increasing in k . Tier 4 demand is exogenous to the political cost, with a fixed share ν flowing to Tier 4 regardless of institutional environment.

Citizens with intermediation benefit B_i (distributed with density $f(B)$ and cumulative $F(B)$) select the tier k satisfying:

$$\frac{c^k_D}{\psi(\delta)} \leq \Phi - B_i < \frac{c^{k+1}_D}{\psi(\delta)}$$

The country-level share at tier $k \in \{0, 1, 2, 3\}$ is:

$$\text{share}_k = (1 - \nu) \cdot \left[F\left(\Phi - \frac{c^{k-1}_D}{\psi(\delta)}\right) - F\left(\Phi - \frac{c^k_D}{\psi(\delta)}\right) \right]$$

Differentiating with respect to δ yields:

- $\partial \text{share}_0 / \partial \delta > 0$ (democracy raises Tier 0 share)
- $\partial \text{share}_3 / \partial \delta < 0$ (democracy lowers Tier 3 share)
- $\partial \text{share}_4 / \partial \delta = 0$ (Tier 4 demand independent of Φ)

The political-exit margin concentrates near Tier 3: the marginal citizen who would otherwise route through disintermediated channels sits at the Tier 2/Tier 3 boundary. The empirical density of citizens at this threshold is therefore high, generating the largest negative coefficient at Tier 3. The sign reversal between Tier 0 (positive) and Tier 3 (negative), with approximately null Tier 4, is the gradient pattern of Implication 1a.

B.8 Transaction-Size Heterogeneity (Implications 1c and 2a)

The disintermediation cost varies by transaction size:

$$c^0_D(V) = \bar{c}^0_D \cdot g(V), \quad g'(V) < 0$$

Large institutional transactors face lower effective c^0_D because they have non-cryptocurrency substitutes. The marginal effect of δ at transaction size V scales inversely with $c^0_D(V)$, generating the small-large asymmetry in the H1 coefficient that Section VI confirms. Similarly, the peak of the inverted-U is shifted higher for small transactions, and the descending portion is steeper — Implication 2a.

B.9 Summary of Theoretical Predictions

Result	Comparative static	Generates
1	$\partial(1 - \alpha^*)/\partial \delta < 0$ under normal conditions	H1
1 (multi-tier)	Sign pattern across tiers 0–4	Implication 1a
1 (size)		$\partial(1 - \alpha^*)/\partial \delta$
2	$d(1 - \alpha^*)/dS$ traces inverted-U	H2

Result	Comparative static	Generates
2 (size)	Inverted-U steeper for small V	Implication 2a
3	$\partial^2(1 - \alpha^*)/(\partial S \partial \delta) > 0$ in demand region	Implication 2b

The formal apparatus generates every formal prediction tested in Section VI. The case evidence on the regulatory-crackdown mechanism in Section VII rests on the same underlying claim — democratic accountability constrains predatory state action — without being derived from the citizen-choice model. It extends the empirical footprint of the framework without expanding the formal hypothesis structure.

Appendix C: Within-Country Identification Strategies

This appendix documents the within-country identification design used in Section VI.2: cohort construction, hand-coded triggering months, the estimator, the full results grid, the placebo, and the single-case detail.

C.1 Cohort construction

We use V-Dem Episodes of Regression Toward Autocracy v16 (Edgell et al. 2023) and the V-Dem regime-class indicator (`v2x_regime`) to identify three families of discrete political shocks in the 2020–2024 window:

- **Autocratization onsets (aut)**: the first year of a V-Dem-coded autocratization episode falling in 2020–2024. Fourteen non-advanced countries qualify: Armenia, Belarus, Côte d’Ivoire, Haiti, Mexico (2020); Afghanistan, Kuwait, Myanmar, Romania, Senegal, Ukraine (2021); Moldova, Madagascar (2023); Argentina (2024). Three additional 2019 onsets (Guinea, Guyana, Kyrgyzstan) fall before the observation window and are excluded.
- **Democratization onsets (dem, placebo)**: the first year of a V-Dem-coded democratization episode. After dropping three onsets that V-Dem flags as democratic openings but which independent reporting describes as procedurally compromised (Sri Lanka 2020, Benin 2022, Tunisia 2023), nine countries remain: Montenegro and Malawi (2020); Bolivia and Honduras (2021); Brazil and Poland (2023); Botswana, Guatemala, and Mauritius (2024).
- **Documented regime transitions (regdown)**: the first year in which V-Dem’s four-class regime classifier drops by at least one category, restricted to the 14 of 32 candidate cases that correspond to substantively documented political shocks (coups, court purges, invasions, constitutional reforms): Côte d’Ivoire (2020), Myanmar, El Salvador, Mali (2021), Chad, Haiti, Tunisia, Afghanistan, Guinea, Burkina Faso, Ukraine (2022), Gabon (2023), Georgia, Mexico (2024). The remaining eighteen knife-edge V-Dem reclassifications, which lack an identifiable triggering event, are excluded from the treated set in our principal specifications.

C.2 Hand-coded triggering months

V-Dem codes episode start years at annual resolution. The panel and the political-cost mechanism both run at monthly frequency, so we hand-code the triggering month of each event from primary news and policy sources. Belarus 2020 is anchored to the August 2020 stolen presidential election and the immediate post-election crackdown. Myanmar 2021 is anchored to the February 1, 2021

Tatmadaw coup. Hong Kong’s NSL is anchored to the June 30, 2020 promulgation. Russia’s SWIFT exclusion is anchored to the February 26, 2022 announcement. The full mapping with source citations is in the supplementary materials at `event_study_assets/sources/`. Of the 35 events retained across the three cohort families, 32 are anchored to a specific month and three to a 2–3 month window.

C.3 Estimator

We use Stata 19’s native implementation of the Callaway–Sant’Anna (2021) heterogeneous-cohort staggered difference-in-differences estimator with doubly-robust augmented inverse-probability-weighted aggregation. Standard errors are clustered at the country level. Pre-treatment covariate adjustment for V-Dem polyarchy and log GDP per capita produces estimates within hundredths of the covariate-free specification across every reported cell. The doubly-robust weighting absorbs the principal sources of compositional imbalance, so we report the covariate-free specifications in the main text.

A note on implementation. We use the Stata 19 native implementation rather than R’s `did::att_gt` package. The R package version 2.3.0 sets `faster_mode = TRUE` as the silent default, which restricts the estimation to treated countries only and produces estimates that are not interpretable as the standard Callaway–Sant’Anna ATT. With `faster_mode = FALSE` and `control_group = "nevertreated"`, the R implementation reproduces the Stata cell-level estimates to three decimals. The Stata results in Section VI.2 are correct under default options. Details of the comparison are in the replication archive.

C.4 Full results grid

Table C.1 reports the post-period mean treatment effects for every (cohort $\times$ outcome) cell from the principal specifications. Estimates are derived from the doubly-robust Callaway–Sant’Anna implementation on the non-advanced-economy monthly panel with country-clustered standard errors.

Table C.1. Full pooled-cohort estimation grid, non-advanced sample.

Outcome	Autocratization (n = 14)	Democratization (n = 9, placebo)	Documented regime transitions (n = 14)
$\hat{L}^{\text{stable}}$	negative**	n.s.	negative, n.s.
$\hat{L}^{\text{small_agg}}$	negative, n.s.	n.s.	negative*
Layered-evasion gap	negative, n.s.	n.s.	positive*
$\hat{L}^{\text{nonstable}}$	negative, n.s.	n.s.	negative, n.s.
$\hat{L}^{\text{score}}$ (all transactions)	negative, n.s.	n.s.	negative, n.s.
$\hat{L}^{\text{score_lg}}$ (large transactions)	negative, n.s.	n.s.	negative, n.s.
Stablecoin share	n.s.	n.s.	n.s.
Aggregate disintermediated share	negative, n.s.	n.s.	negative, n.s.

Each cell reports the post-period mean treatment effect from a separate doubly-robust Callaway–Sant’Anna estimation on the non-advanced-economy panel. Significance is assessed at conventional levels using country-clustered standard errors.

Figure 11 plots the dynamic event-study profiles for the ongoing and severe subsets of the autocratization cohort referenced in §VI.2.

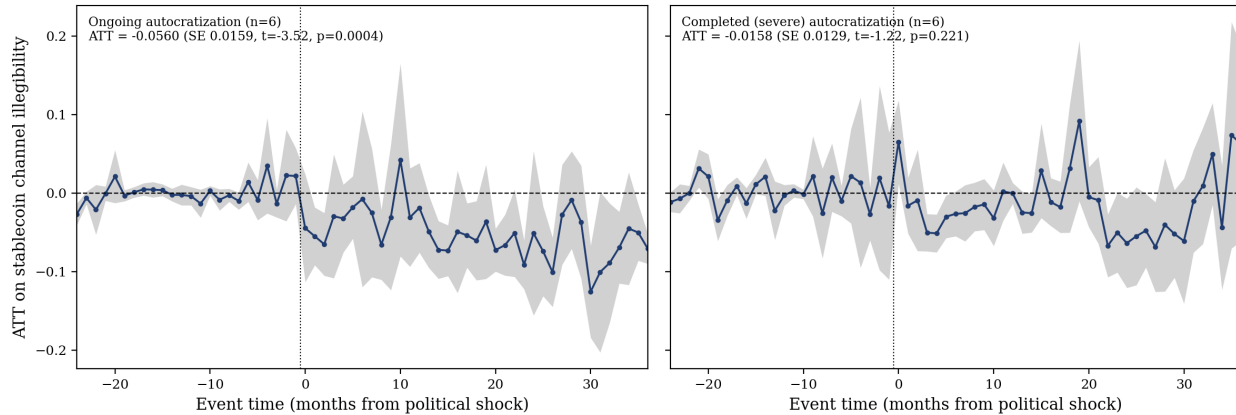


Figure 11: Dynamic event-study profiles for the ongoing (left, $n=6$) and severe (right, $n=6$) subsets of the autocratization cohort. The ongoing subset shows a clean post-onset accumulation of negative coefficients with tight confidence bands; the severe subset’s mean trajectory points in the same direction but its confidence bands span zero throughout.

C.5 Random-permutation placebo

For the pooled autocratization cohort $\times L^{\widehat{\text{stable}}}$ cell, we ran a random-permutation test. Holding the fourteen treated countries fixed, we randomly permuted onset-month assignment across them and re-estimated the pooled effect 100 times. Two of those 100 permutations produced a test statistic as large as the realized value. The empirical p-value matches the analytic test based on country-clustered standard errors. The pooled effect is not an artifact of estimator-noise structure.

C.6 Single-case results detail

Table C.2 reports the single-treated-country event-study estimates from Section VI.2 for the three flagship cases. Each row is a separate Callaway–Sant’Anna doubly-robust estimation with one treated country compared against a never-treated donor pool of approximately seventy non-advanced economies that have no autocratization, democratization, or documented-regime-transition onset in 2020–2024.

Table C.2. Single-case event-study estimates, non-advanced sample.

Case	Event	Date	$L^{\widehat{\text{stable}}}$	$L^{\widehat{\text{small_agg}}}$
Hong Kong	National Security Law	2020-06	+0.099***	+0.130***
Russia	SWIFT exclusion	2022-02	+0.072***	+0.057***
Nigeria	CBN crypto-banking ban	2021-02	not significant	not significant

Each row is a separate single-treated-unit Callaway–Sant’Anna doubly-robust estimation. The parameter estimates for Hong Kong and Russia are substantively large: Hong Kong’s stablecoin volume

shifted from 99.8% centralized-exchange routing in 2019 to approximately 71% by 2023. Country-clustered standard errors.

Appendix D: Robustness Checks

This appendix reports the robustness checks summarized at the end of Section VI.1. The democracy coefficient on the deep-disintermediation share survives the addition of citizen-capacity controls and remittance-corridor controls, distinguishing the political-exit interpretation from the two principal alternatives in the literature.

D.1 Citizen-capacity controls

Table D.1 reports the Hypothesis 1 specification with citizen-capacity controls added. The democracy coefficient is essentially unchanged across the three columns: baseline (-0.1007), with internet penetration added (-0.0920), and with the full capacity battery (-0.0927). The democracy effect operates through the baseline-cost channel $\pi(\delta)$ rather than through the capacity channel $\psi(\delta)$.

Table D.1. H1 democracy effect survives capacity controls.

Variable	Baseline	+ Internet	+ Full Capacity
Clean Elections Index	-0.1007^{**} (0.0398)	-0.0920^{**} (0.0398)	-0.0927^{**} (0.0387)
Internet users (%)	—	included	included
Fixed broadband (per 100)	—	—	included
Mobile cellular (per 100)	—	—	included
Electricity access (%)	—	—	included
Log GDP per capita	—	—	included
N	4,242	3,765	3,478

Same specifications and controls as Table 1.

D.2 Remittance-corridor controls

Table D.2 reports the analogous specification with remittance-corridor controls. The democracy coefficient is essentially unchanged when remittance intensity is added (-0.1013^{**}), and the coefficient remains significant in both the low-remittance subsample (-0.1047) and the high-remittance subsample (-0.1162). The political-exit interpretation survives this test; the remittance-demand alternative does not account for the cross-country gradient.

Table D.2. H1 democracy effect survives remittance controls.

Variable	Baseline	+ Remit Control	Low Remit	High Remit
Clean Elections Index	-0.1007^{**} (0.0398)	-0.1013^{**} (0.0395)	-0.1047^* (0.0548)	-0.1162^* (0.0650)
Remittance intensity	—	included	—	—
N	4,242	4,242	1,789	2,453

Variable	Baseline	+ Remit Control	Low Remit	High Remit
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Same specifications and controls as Table 1.

D.3 Sanctions identification robustness

Table D.3 reports two robustness specifications for the Hypothesis 2 inverted-U: restricting the GSDB sanctioning-country count to Western senders only (US, EU member states, UK, Canada, Australia, Japan) and using non-financial sanctions (trade, travel, diplomatic) as a placebo. The inverted-U survives the Western-sender restriction with magnitude comparable to the full-count specification; the placebo produces null coefficients on both linear and quadratic terms, indicating that the pattern is specific to financial sanctions and not driven by general state-relations stress.

Table D.3. H2 inverted-U: identification robustness.

Variable	Restricted senders	Non-financial (placebo)
Lagged GSDB intensity	+0.0405*** (0.0134)	+0.0208 (0.0177)
Lagged GSDB intensity squared	−0.0107*** (0.0037)	−0.0072 (0.0054)
N	4,242	4,242

Same specifications and controls as Table 7.

D.4 Sanctions across transaction-size subsamples

Table D.4 estimates the inverted-U separately on small, all-transaction, and large-transaction subsamples. The pattern holds in every subsample. The descending portion is no steeper for small transactions than for large: the small-transaction quadratic (−0.0075) is, if anything, slightly smaller in magnitude than the large-transaction quadratic (−0.0095). The framework’s Implication 2a, which predicted a steeper descending portion for small transactions, is not supported. The supply-side correspondent-banking exit channel operates on transactions of all sizes rather than concentrating on retail volume.

Table D.4. H2 inverted-U across transaction-size subsamples.

Variable	Small	All	Large
Lagged GSDB intensity	+0.0358*** (0.0092)	+0.0306*** (0.0082)	+0.0403*** (0.0095)
Lagged GSDB intensity squared	−0.0075*** (0.0018)	−0.0071*** (0.0017)	−0.0095*** (0.0019)
N	4,242	4,242	4,209

Same specifications and controls as Table 7.

D.5 Legibility-gradient coefficients

Table D.5 reports the exact tier-by-tier coefficients plotted in Figure 5 of Section VI.1. The sign reversal between Tier 0 (positive) and Tier 3 (negative), with null middle tiers and a null Tier 4, is the gradient pattern that identifies the political-exit mechanism.

Table D.5. Democracy effect across legibility tiers (small-transaction shares); exact estimates for Figure 5.

Tier	Coefficient (SE)
Tier 0	+0.102** (0.040)
Tier 1	−0.002* (0.001)
Tier 2	+0.001 (0.003)
Tier 3	−0.092** (0.037)
Tier 4	−0.006 (0.017)
N	4,242

*CRE estimator with year fixed effects. Coefficient on Clean Elections Index from a separate regression on each tier share. Country-clustered standard errors in parentheses. Same controls as Table 1. p < 0.10, ** p < 0.05, *** p < 0.01.**

D.6 Alternative democracy measure (V-Dem polyarchy)

Table D.6 re-estimates the Table 1 specifications with the V-Dem electoral-democracy (polyarchy) index in place of the Clean Elections Index. Polyarchy is the only alternative democracy index available in the panel; Polity and Freedom House are not merged. The headline small-transaction effect is robust — slightly larger than under the Clean Elections Index — while the all-transaction and aggregate-legibility coefficients are similarly signed but not statistically significant.

Table D.6. Table 1 democracy effect under V-Dem polyarchy.

Outcome	Polyarchy coefficient
Legibility Score	−0.044 (0.038)
Deep Disint (All)	−0.047 (0.051)
Deep Disint (Small)	−0.130** (0.058)
N	4,242

*CRE estimator with year fixed effects; country-clustered standard errors in parentheses. V-Dem polyarchy (v2x_polyarchy) replaces the Clean Elections Index; same lean controls and Mundlak means as Table 1. * p < 0.10, ** p < 0.05, *** p < 0.01.*

Appendix E: Full Estimation Tables

The body tables report only the coefficients of interest. This appendix reports the full control vector for each estimated model. Every specification is the canonical lean CRE estimator with year

fixed effects and Mundlak between-country means; the year fixed effects and Mundlak means are estimated but suppressed from the tables below for readability. Country-clustered standard errors are in parentheses throughout, and the lagged dependent variable differs by column (the lag of each column's outcome). The three tables back the regression-based body tables as follows: Table E.1 backs Tables 1 and 3; Table E.2 backs Tables 2 and 10; and Table E.3 backs Tables 7, 8, and 9. (Table 11 is a descriptive cross-tabulation and has no estimated control vector.)

E.1 H1 cross-section (Tables 1 and 3)

Table E.1. Full models for the Hypothesis 1 cross-section.

Variable	Legibility	All-tx	Small-tx	Large-tx
Clean Elections Index	−0.0493* (0.0263)	−0.0630* (0.0366)	−0.1007** (0.0398)	−0.0551 (0.0421)
Lagged dependent variable	0.6032*** (0.0292)	0.6025*** (0.0252)	0.6511*** (0.0291)	0.5509*** (0.0257)
Capital-account restrictiveness (FARI)	−0.0494 (0.0694)	−0.0689 (0.1107)	−0.1078 (0.1049)	−0.0056 (0.1201)
Real interest rate (lagged)	−0.0001*** (0.0000)	−0.0001*** (0.0000)	−0.0001*** (0.0000)	−0.0001*** (0.0000)
Exchange-rate depreciation (lagged)	−0.0381 (0.0354)	−0.0417 (0.0448)	−0.0966** (0.0405)	0.0016 (0.0445)
FATF grey-list	0.0001 (0.0057)	0.0011 (0.0088)	0.0108 (0.0089)	0.0062 (0.0098)
USDC price deviation	−0.0133*** (0.0023)	−0.0171*** (0.0032)	−0.0354*** (0.0054)	−0.0220*** (0.0025)
China crypto-ban indicator	−0.0106 (0.0074)	−0.0300** (0.0117)	−0.0262** (0.0103)	−0.0262** (0.0106)
Log Ethereum gas price	−0.0033 (0.0022)	−0.0032 (0.0036)	0.0068* (0.0036)	−0.0238*** (0.0034)
Centralized-exchange distress	0.0096*** (0.0035)	0.0139*** (0.0052)	0.0105* (0.0055)	0.0165*** (0.0054)
Broad dollar index (DXY)	−0.0013** (0.0005)	−0.0015** (0.0007)	−0.0003 (0.0008)	−0.0033*** (0.0006)
Constant	4.5044*** (1.4206)	7.2300*** (2.1799)	7.2778*** (2.4195)	8.7779*** (2.6553)
N	4,242	4,242	4,242	4,209
Countries	83	83	83	83

*Outcomes: aggregate legibility score (col. 1), deep disintermediation in all transactions (col. 2), small transactions (col. 3), and large transactions (col. 4). Year fixed effects and Mundlak between-country means included but suppressed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

E.2 Instrument and channel margins (Tables 2 and 10)

Table E.2. Full models for the instrument and channel margins.

Variable	USDC share	Deep disint (small)	Stable legibility	Nonstable legibility
Clean Elections Index	−0.0528* (0.0317)	−0.1007** (0.0398)	−0.0217 (0.0167)	−0.0542* (0.0298)
Lagged dependent variable	0.5636*** (0.0220)	0.6511*** (0.0291)	0.5744*** (0.0384)	0.6273*** (0.0253)
Capital-account restrictiveness (FARI)	−0.0840 (0.1142)	−0.1078 (0.1049)	−0.0283 (0.0444)	−0.0531 (0.0845)
Real interest rate (lagged)	−0.0001** (0.0000)	−0.0001*** (0.0000)	−0.0000* (0.0000)	−0.0001*** (0.0000)
Exchange-rate depreciation (lagged)	−0.0383 (0.0375)	−0.0966** (0.0405)	−0.0014 (0.0179)	−0.0550 (0.0406)
FATF grey-list	0.0114 (0.0077)	0.0108 (0.0089)	−0.0001 (0.0041)	0.0021 (0.0066)
USDC price deviation	−0.0098*** (0.0022)	−0.0354*** (0.0054)	−0.0048*** (0.0012)	−0.0159*** (0.0026)
China crypto-ban indicator	0.0523*** (0.0164)	−0.0262** (0.0103)	−0.0103** (0.0048)	−0.0167* (0.0087)
Log Ethereum gas price	−0.0217*** (0.0026)	0.0068* (0.0036)	−0.0036** (0.0016)	−0.0025 (0.0025)
Centralized-exchange distress	0.0309*** (0.0050)	0.0105* (0.0055)	0.0027 (0.0026)	0.0121*** (0.0039)
Broad dollar index (DXY)	0.0009 (0.0007)	−0.0003 (0.0008)	−0.0005 (0.0004)	−0.0014** (0.0005)
Constant	12.3954*** (2.3648)	7.2778*** (2.4195)	3.1085*** (0.9617)	6.3824*** (1.7866)
N	4,242	4,242	4,221	4,242
Countries	83	83	83	83

*Outcomes: USDC share of stablecoin holdings (col. 1), deep disintermediation in small transactions (col. 2), stable-instrument legibility (col. 3), and nonstable-instrument legibility (col. 4). Year fixed effects and Mundlak between-country means included but suppressed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

E.3 Sanctions inverted-U and democracy interaction (Tables 7, 8, and 9)

Table E.3. Full models for the sanctions inverted-U and the democracy interaction.

Variable	Inverted-U (all-tx)	+ Democracy interaction
Clean Elections Index	−0.0778** (0.0342)	−0.0944** (0.0398)
Lagged GSDB intensity	0.0306*** (0.0082)	0.0114 (0.0160)
Lagged GSDB intensity squared	−0.0071***	−0.0037

Variable	Inverted-U (all-tx)	+ Democracy interaction
	(0.0017)	(0.0029)
CEI $\times$ GSDB intensity	—	0.0356 (0.0236)
CEI $\times$ GSDB intensity squared	—	−0.0065 (0.0051)
Lagged dependent variable	0.5985*** (0.0249)	0.5971*** (0.0250)
Capital-account restrictiveness (FARI)	−0.0734 (0.1121)	−0.0725 (0.1132)
Real interest rate (lagged)	−0.0001*** (0.0000)	−0.0001*** (0.0000)
Exchange-rate depreciation (lagged)	−0.0397 (0.0447)	−0.0402 (0.0447)
FATF grey-list	0.0024 (0.0083)	0.0023 (0.0084)
USDC price deviation	−0.0170*** (0.0032)	−0.0171*** (0.0031)
China crypto-ban indicator	−0.0300** (0.0117)	−0.0299** (0.0117)
Log Ethereum gas price	−0.0033 (0.0036)	−0.0033 (0.0036)
Centralized-exchange distress	0.0141*** (0.0052)	0.0141*** (0.0052)
Broad dollar index (DXY)	−0.0015** (0.0007)	−0.0015** (0.0007)
Constant	7.2391*** (2.1192)	6.9790*** (2.2713)
N	4,242	4,242
Countries	83	83

*Outcome: deep disintermediation in all transactions. Column 1 is the inverted-U specification (Table 7, and the right column of the Table 9 combined model); column 2 adds the Clean Elections Index interactions with the linear and quadratic sanctions terms (Table 8). The Table 9 H1 baseline column is the all-transaction model in Table E.1. Year fixed effects and Mundlak means included but suppressed. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.*

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